Q1.

ANS

Q. What is the purpose of using keyword-based query? Briefly explain any 3 types of keyword-based queries.

(10 Marks)

1) Introduction :

Keyword-based query is one of the most common and widely used query techniques in Information Retrieval (IR). In this approaach, users express their information need by using a set of keywords (single words or phrases) rather than a formal query language. Search engines and IR systems match these keywords with the terms present in the documents and return a ranked list of relevant documents.

2) Purpose of using keyword-based query :

The main purposes are :

① **Simplicity and Ease of Use :** Users can easily enter natural words without learning any special query syntax or language.

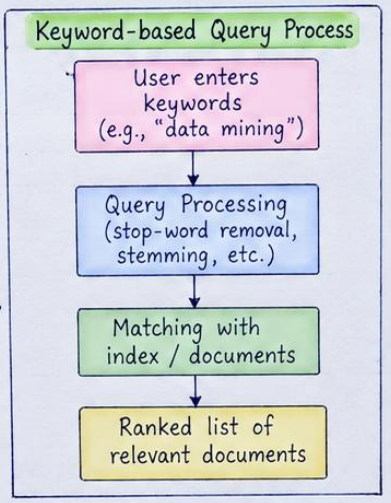
② **Quick Information Access :** It allows fast retrieval of relevant documents from large collections.

③ **Supports Natural Language :** Users can use familiar words as they think, making it user friendly.

④ **Wide Applicability :** It is used in web search engines, digital libraries, online databases, and many IR systems.

⑤ **Effective for General Queries :** Works well for general information needs where exact structured queries are not required.

⑥ **Basis for Advanced Features :** It can be extended with techniques like stemming, stop-word removal, and ranking to improve retrieval performance.



3) Types of Keyword-based Queries :

There are different types of keyword-based queries used in IR systems. Any three are explained below :

① Single Keyword Query (Simple Query)

- It consists of a single keyword.
- The system retrieves all documents containing that keyword.
- Useful for general and broad searches.

Example :

Query : "python"

Retrieves documents related to python programming, python language, etc.

② Multi-keyword Query (Conjunctive/Disjunctive)

- It consists of two or more keywords combined using Boolean operators AND, OR, NOT.
- Helps to refine the search and get more precise results.

Example :

- Conjunctive (AND) : "data AND mining" → documents containing both words.
- Disjunctive (OR) : "java OR python" → documents containing either word.
- Exclusion (NOT) : "apple NOT fruit" → documents about apple company, not fruit.

③ Phrase Query (Exact Match Query)

- It searches for an exact sequence of keywords (a phrase).
- Useful when the user wants documents containing the exact phrase in the same order.
- More specific and reduces irrelevant results.

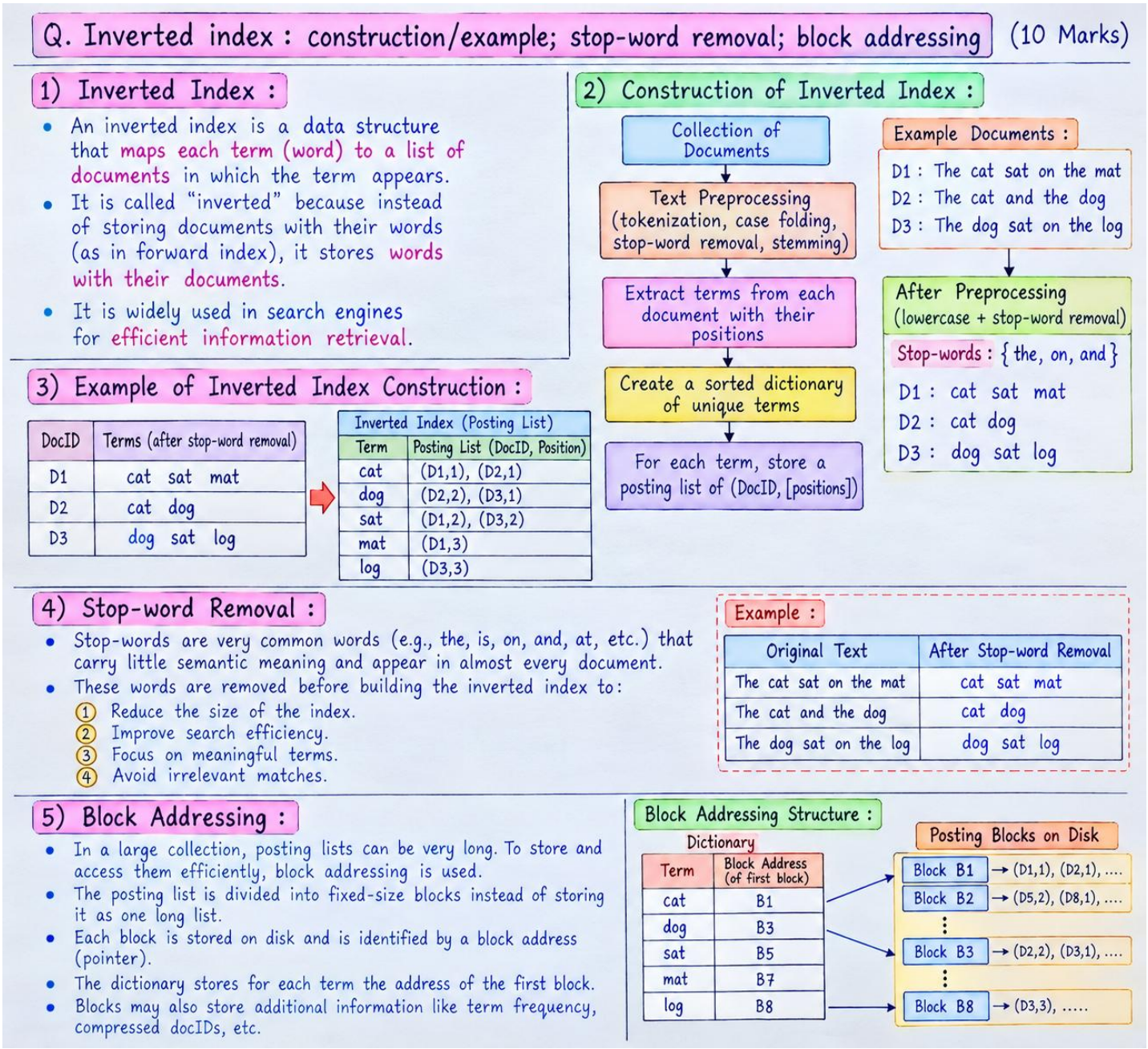
Example :

Query : "information retrieval"

Retrieves only those documents where the exact phrase "information retrieval" appears together.

Q2.

ANS


Q. Inverted index : construction/example; stop-word removal; block addressing (10 Marks)
1) Inverted Index :
An inverted index is a data structure that maps each term (word) to a list of documents in which the term appears.
It is called "inverted" because instead of storing documents with their words (as in forward index), it stores words with their documents.
It is widely used in search engines for efficient information retrieval.
2) Construction of Inverted Index :
Collection of Documents
Text Preprocessing (tokenization, case folding, stop-word removal, stemming)
Extract terms from each document with their positions
Create a sorted dictionary of unique terms
For each term, store a posting list of (DocID, [positions])
Example Documents :
D1 : The cat sat on the mat
D2 : The cat and the dog
D3 : The dog sat on the log
After Preprocessing (lowercase + stop-word removal)
Stop-words : { the, on, and }
D1 : cat sat mat
D2 : cat dog
D3 : dog sat log
3) Example of Inverted Index Construction :
DocID | Terms (after stop-word removal)
D1 | cat sat mat
D2 | cat dog
D3 | dog sat log
Inverted Index (Posting List)
Term | Posting List (DocID, Position)
cat | (D1,1), (D2,1)
dog | (D2,2), (D3,1)
sat | (D1,2), (D3,2)
mat | (D1,3)
log | (D3,3)
4) Stop-word Removal :
Stop-words are very common words (e.g., the, is, on, and, at, etc.) that carry little semantic meaning and appear in almost every document.
These words are removed before building the inverted index to:
1 Reduce the size of the index.
2 Improve search efficiency.
3 Focus on meaningful terms.
4 Avoid irrelevant matches.
Example :
Original Text | After Stop-word Removal
The cat sat on the mat | cat sat mat
The cat and the dog | cat dog
The dog sat on the log | dog sat log
5) Block Addressing :
In a large collection, posting lists can be very long. To store and access them efficiently, block addressing is used.
The posting list is divided into fixed-size blocks instead of storing it as one long list.
Each block is stored on disk and is identified by a block address (pointer).
The dictionary stores for each term the address of the first block.
Blocks may also store additional information like term frequency, compressed docIDs, etc.
Block Addressing Structure :
Dictionary
Term | Block Address (of first block)
cat | B1
dog | B3
sat | B5
mat | B7
log | B8
Posting Blocks on Disk
Block B1 → (D1,1), (D2,1), ...
Block B2 → (D5,2), (D8,1), ...
Block B3 → (D2,2), (D3,1), ...
Block B8 → (D3,3), ...

Q3.

ANS

Q. Multimedia IR / indexing / searching (including image and video IR) (10 Marks)

1) Introduction :

- Multimedia Information Retrieval (MMIR) is the process of searching, indexing and retrieving multimedia data such as images, audio and videos from large collections.
- It extends traditional text-based IR by handling different data types and their unique features (visual, audio, temporal).
- The goal is to efficiently find relevant multimedia objects based on user queries (e.g., image, text, or video).

2) Indexing in Multimedia IR :

Indexing converts multimedia content into compact representations (features/metadata) so that efficient searching can be performed.

Steps in Indexing :

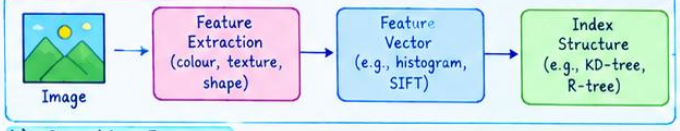
1. Feature Extraction – Extract important features from the media (e.g., colour, texture, shape for images; keyframes, motion, audio features for videos).
2. Feature Representation – Represent features using vectors or descriptors (e.g., colour histogram, SIFT, MPEG-7).
3. Creating Index – Store the feature vectors in an index structure (e.g., inverted index, feature vector index).
4. Metadata Storage – Store additional info like title, tags, caption, duration, etc.

3) Image IR :

a) Indexing of Images :

- Images are represented using visual features such as colour, texture, shape and spatial layout.
- Common features: Colour histogram, texture descriptors (GLCM), shape descriptors, SIFT, SURF.
- Features are stored in a feature vector Index (e.g. KD-tree, R-tree, or inverted index for visual words).

Image Indexing Process :



b) Searching Images :

- User submits a query image or example image.
- System extracts features and compares with indexed images using similarity measures (e.g., Euclidean distance, cosine similarity).
- Returns top-k most similar images.

Example – Image Search :

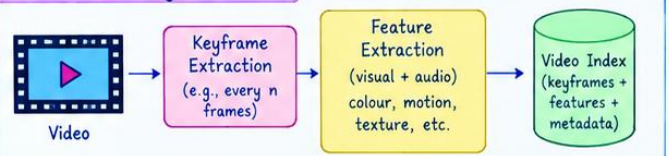


4) Video IR :

a) Indexing of Videos :

- Videos are indexed using keyframes, motion information, visual features and audio features.
- The video is divided into keyframes (frames at certain intervals).
- Features such as colour, texture, motion vectors, and audio are extracted and stored in an index.

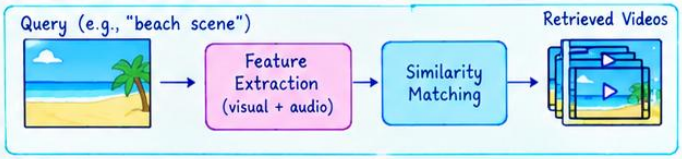
Video Indexing Process :



b) Searching Videos :

- User provides a text query, example frame or video clip.
- System extracts features (text, visual or audio) and matches with indexed features using similarity measures.
- Returns relevant video clips or keyframes.

Example – Video Search :



5) Advantages of Multimedia IR :

- Provides rich and intuitive search experience (images/videos).
- Combines multiple features (visual, audio, text).
- Useful for applications like surveillance, e-commerce, social media, content recommendation, etc.

6) Conclusion :

Multimedia IR uses indexing and advanced feature extraction techniques to enable efficient searching of images and videos, helping users find relevant multimedia content quickly and accurately.

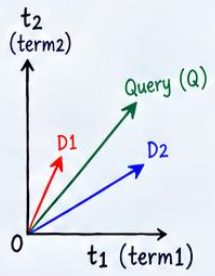
Q4.

ANS

Q. Vector-space model / ranking / TF-IDF / similarity (10 Marks)

1) Vector-space model :

- Vector-space model (VSM) represents each document and query as a vector of term weights in a multi-dimensional space, where each dimension corresponds to a term from the vocabulary.
- Each document is a point (vector) in this space.
- Documents that are closer to the query vector (i.e., have higher similarity) are considered more relevant and are ranked higher.
- It uses term weights (commonly TF-IDF) and cosine similarity for measuring similarity.



Documents closer to the query vector are more similar and ranked higher.

2) Term Frequency – Inverse Document Frequency (TF-IDF) :

- TF-IDF is a widely used term weighting scheme in VSM.
- It gives high weight to terms that are frequent in a document but rare across the collection.

Term Frequency (TF) :

$$TF(t, d) = \frac{\text{Number of times term t appears in document d}}{\text{Total number of terms in document d}}$$

Inverse Document Frequency (IDF) :

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

N = total number of documents
df(t) = number of documents containing term t

TF – IDF Weight : $w(t, d) = TF(t, d) \times IDF(t)$

3) Example :

Consider the following 3 documents and a query.

Document	Text
D1	the cat sat on the mat
D2	the cat and the dog
D3	the dog sat on the log
Query (Q)	cat sat

Vocabulary (unique terms) :

{ cat, sat, mat, dog, log }
Total documents (N) = 3

Document Frequency (df) :

df(cat) = 2 df(mat) = 1
df(sat) = 2 df(log) = 1
df(dog) = 2

4) Compute IDF values :

Term	df(t)	IDF(t) = log (3/df(t))
cat	2	0.1761
sat	2	0.1761
mat	1	0.4771
dog	2	0.1761
log	1	0.4771

Using log base 10 :

$IDF(cat) = \log\left(\frac{3}{2}\right) = 0.1761$

$IDF(mat) = \log\left(\frac{3}{1}\right) = 0.4771$

(same for log)

5) Compute TF-IDF weights :

Term	D1 (the cat sat on the mat)	D2 (the cat and the dog)	D3 (the dog sat on the log)	Q (cat sat)
cat	0.0352	0.0352	0	0.0880
sat	0.0352	0	0.0352	0.0880
mat	0.0954	0	0	0
dog	0	0.0352	0.0352	0
log	0	0	0.0954	0

TF-IDF weight = TF x IDF

Example for D1, term "cat": $TF = \frac{1}{6} = 0.1667$

Weight = 0.1667 x 0.1761 = 0.0352

6) Similarity using Cosine Similarity :

- Cosine similarity between query Q and document D:

$$Sim(Q, D) = \frac{\sum w_i \times w_j}{\sqrt{\sum w_i^2} \times \sqrt{\sum w_j^2}}$$

where w_i = weight of term i in Q,
w_j = weight of term i in D.

Document	Dot Product (Q·D)	|Q|	|D|	Cosine Similarity
D1	0.00248	0.1245	0.1043	0.190
D2	0.00124	0.1245	0.0498	0.200
D3	0.00124	0.1245	0.1043	0.096

7) Ranking of Documents :

- Documents are ranked in decreasing order of their similarity with the query.
- From the above example :

Rank	Document	Similarity
1	D2	0.200
2	D1	0.190
3	D3	0.096

Hence, the most relevant document to the query "cat sat" is D2, followed by D1 and D3.

Q5.

ANS

Q. User relevance feedback : Definition, process, types/approaches, and contribution to result quality (10 Marks)

1) Definition :

- User relevance feedback (URF) is a technique in Information Retrieval (IR) where the user provides feedback on the relevance of retrieved documents to refine the search results.
- It uses the user's judgments (relevant / non-relevant) to modify the query and retrieve more accurate and useful results in the next iteration.

2) Process of User Relevance Feedback :

1. User submits initial query
   - User enters a query (e.g., "machine learning").
2. Retrieve initial set of documents
   - System retrieves a ranked list of documents using IR model (e.g., Vector Space Model).
3. User provides relevance feedback (mark relevant / non-relevant)
   - User marks some documents as relevant (R) and some as non-relevant (NR).
4. Modify/expand query using feedback
   - System modifies the query using the feedback (e.g., Relevance Feedback Algorithm such as Rocchio) by increasing weights of terms from relevant documents and decreasing weights of terms from non-relevant documents.
5. Retrieve improved results
   - System retrieves a new ranked list of documents which are more relevant to the user's need.
6. Repeat (if needed) until satisfactory results are obtained
   - Steps 3-5 can be repeated for better results.

Example :

Initial Query : "neural networks"
User marks relevant docs : d1, d3
User marks non-relevant docs : d2

→ System expands/adjusts query using important terms from d1, d3 and reduces effect of terms from d2 → retrieves improved results.

3) Types / Approaches of User Relevance Feedback :

A. Explicit Relevance Feedback

- User explicitly marks documents as relevant or non-relevant (e.g., by clicking "relevant" / "not relevant").
- Feedback is directly used to update the query.

Example : Google's "relevant" feedback, manual selection.

B. Implicit Relevance Feedback

- System automatically infers relevance from the user's behaviour without explicit input.
- Uses actions like :
  - Click-through data
  - Time spent on a document
  - Document downloads
  - Scroll behaviour

Example : Search engine analysing clicked results.

4) Techniques / Models used for Relevance Feedback :

Technique / Model	Description	Key Idea
Rocchio Algorithm (Vector Space Model)	Most widely used relevance feedback model.	Modifies query vector using relevant and non-relevant documents.
Query Expansion	Adds new relevant terms to the original query.	Uses terms from relevant documents.
Term Re-weighting	Increases weight of relevant terms and decreases weight of non-relevant terms.	Adjusts term weights in query vector.
Probabilistic Feedback	Uses probabilistic models to estimate term relevance.	Updates query based on probability of relevance.
Language Model based Feedback	Uses language models to refine query.	Adjusts query using relevant document model.

5) Contribution to Result Quality :

1. Improves Precision : Retrieves more relevant documents by learning from user feedback.
2. Improves Recall : Helps in discovering additional relevant documents that were missed in the initial search.
3. Reduces Query Ambiguity : Clarifies user's actual information need using feedback.
4. Personalizes Search Results : Adapts to individual user preferences and interests.
5. Enhances Ranking : Re-ranks documents based on user feedback, providing more useful and accurate results.
6. Increases User Satisfaction : Users get results that are closer to their needs with fewer irrelevant documents.

Q6.

ANS

Q. Objectives of IRS / Information Retrieval Process (10 Marks)

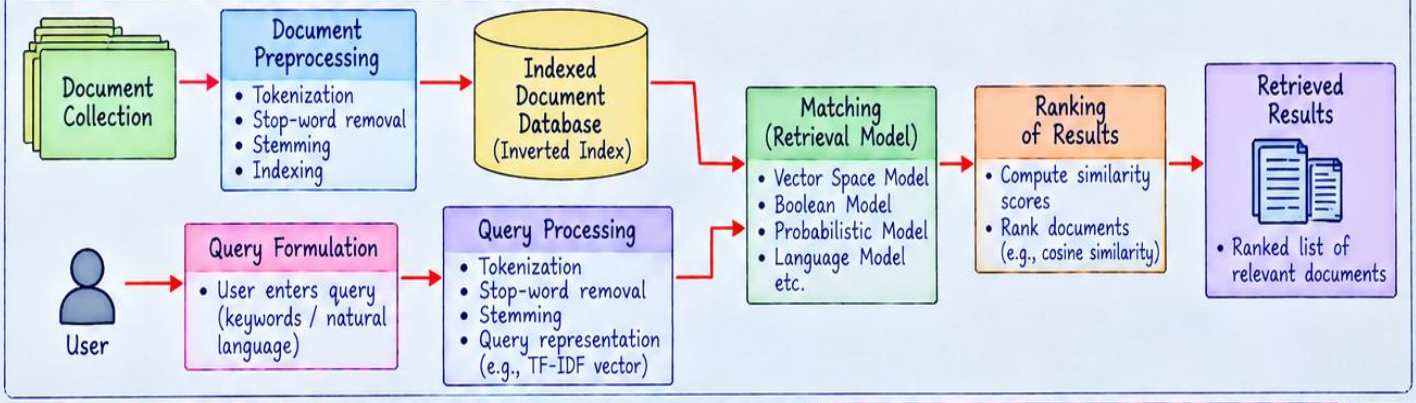
1) Introduction :

- Information Retrieval System (IRS) is a system that stores, organizes and provides access to large collections of unstructured or semi-structured data (mainly text, but also multimedia).
- The main goal is to **retrieve relevant information** that satisfies the user's information need from a large collection of documents.

2) Objectives of Information Retrieval System :

1. **Retrieve Relevant Information :** Return documents that are most relevant to the user's query.
2. **High Precision and Recall :** Improve the accuracy of retrieved results (relevant docs) and minimize irrelevant docs.
3. **Efficient Search :** Handle large collections and provide fast retrieval.
4. **User Satisfaction :** Understand user's information need and deliver useful and meaningful results.
5. **Support Various Data Types :** Handle text, images, audio, video etc.
6. **Scalability and Flexibility :** Adapt to growing data and different user requirements.

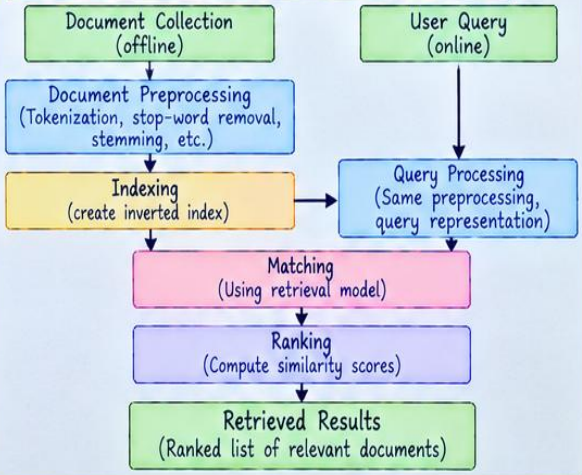
3) Information Retrieval Process (Steps) :



4) Steps in Detail :

1. **Document Collection :** Gather and store documents from various sources (web pages, digital libraries, databases, etc.).
2. **Document Preprocessing :** Clean and normalize the text (tokenization, stop-word removal, stemming/lemmatization) and create an index.
3. **Indexing :** Build an inverted index to store terms and their document references for efficient retrieval.
4. **Query Formulation :** User enters a query in keywords or natural language.
5. **Query Processing :** Apply preprocessing on the query and represent it using the same method as documents (e.g., TF-IDF vector).
6. **Matching / Retrieval :** Compare the query with document representations using a retrieval model (e.g., vector space model) to find relevant documents.
7. **Ranking :** Calculate similarity scores and rank documents in decreasing order of relevance.
8. **Output / Presentation :** Display the ranked list of relevant documents to the user (with snippets, titles, etc.).

5) Retrieval Process - Flow Diagram :



Q7.

ANS

Q. Natural-language modelling : issues and solutions. Ambiguity, synonymy, morphology, context and possible solutions.

(10 Marks)

1) Introduction :

- Natural language modelling (NLM) deals with representing and processing human language in Information Retrieval (IR).
- Human language is flexible and expressive but also complex and ambiguous, which creates challenges for IR systems.
- The goal of NLM is to bridge the gap between user's natural language queries and the structured representation of documents so that relevant documents can be retrieved effectively.

2) Issues in Natural-Language Modelling :

Human language has several characteristics that make retrieval difficult. Major issues are :

1. Ambiguity
2. Synonymy
3. Morphology
4. Context

These issues lead to mismatch between user's query and document representation, resulting in poor retrieval performance.

3) Ambiguity :

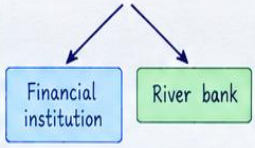
- Ambiguity means a word or a query can have multiple meanings depending on the context.
- It can be of two types :

1) Lexical Ambiguity

A word has multiple meanings.

Example :

- "bank"

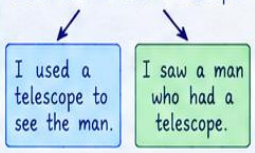


2) Syntactic Ambiguity

Same sentence can have different grammatical interpretations.

Example :

"I saw the man with a telescope."



Solutions :

- Use word sense disambiguation (WordNet, semantic networks).
- Use context information from surrounding words.
- Use probabilistic models (e.g., language models).
- Use user feedback.

4) Synonymy :

- Different words can have similar meaning (synonyms), which may cause relevant documents to be missed if only exact word matching is used.

Example :

- "car" → "automobile"
- "student" → "learner"
- "big" → "large"

Solutions :

- Use thesaurus or lexical resources (e.g., WordNet).
- Query expansion with synonyms.
- Use semantic similarity measures (word embeddings).
- Use relevance feedback to learn synonymous terms.

5) Morphology :

- Words can appear in different morphological forms (inflection, derivation), but have same root meaning.
- IR systems may treat them as different terms, reducing recall.

Example :

- "compute", "computes", "computed", "computing" → same root word "comput"

Solutions :

- Stemming (e.g., Porter Stemmer) → remove suffixes.
- Lemmatization → convert word to its base form using linguistic rules.
- Use morphological analysis.
- Use n-gram models for handling variants.

6) Context :

- The meaning of a word or query depends on its context (surrounding words, domain, user intention, previous queries).
- Without considering context, the system may retrieve irrelevant documents.

Example :

"jaguar speed" {
- Jaguar (animal) → speed of animal
- Jaguar (car brand) → speed of car

Solutions :

- Use contextual language models (e.g., BERT).
- Use phrase queries and proximity search.
- Use user profile, search history and domain information.
- Apply query expansion with context-based terms.
- Use machine learning and deep learning models.

7) Summary and Impact on IR :

Issue	Problem	Possible Solutions
Ambiguity	Multiple meanings of words/sentences	Word sense disambiguation, context, probabilistic models
Synonymy	Different words with same meaning	Thesaurus (WordNet), query expansion, word embeddings
Morphology	Different, word forms	Stemming, lemmatization, morphological analysis
Context	Meaning depends on surrounding words	Contextual models (BERT), user profile, phrase queries

Result :

- → Improves precision and recall.
- → Retrieves more relevant documents.
- → Provides better understanding of user's information need.
- → Enhances overall search quality in Information Retrieval systems.

Q8.

ANS

Q. Search engine architecture / purpose / tasks / retrieval process. Prepare architecture and operation stages with a neat diagram. (10 Marks)

1) Purpose of Search Engine :

- To search and retrieve relevant information from a large collection of web pages based on user queries.
- To provide fast, accurate and useful results to the users.
- To organize, store and index huge amount of data from the web.
- To help users find relevant documents, images, videos, news, etc.

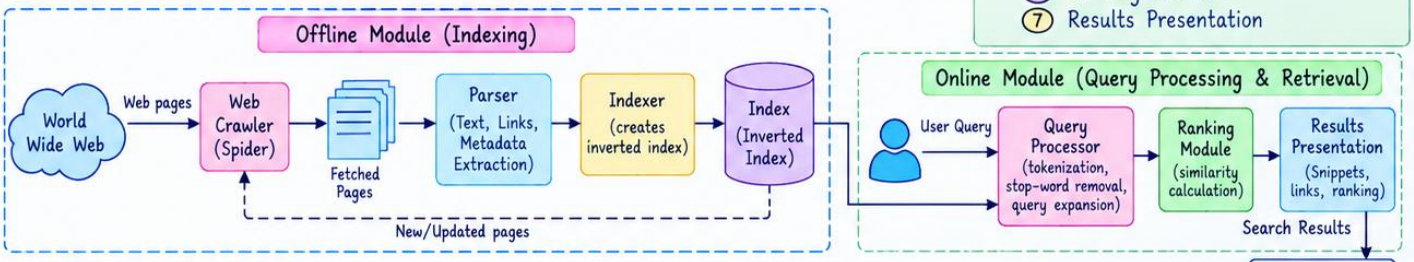
2) Tasks of a Search Engine :

- Crawling: discover and fetch web pages from the Internet.
- Parsing: extract text, links and metadata from fetched pages.
- Indexing: store and organize the extracted information in an index.
- Query Processing: interpret user query (tokenization, stop-word removal, etc.).
- Searching: match query with index and retrieve relevant documents.
- Ranking: order the retrieved documents based on relevance.
- Result Presentation: display results to user with snippets, highlights, etc.

3) Search Engine Architecture :

- A search engine consists of two main modules:
  (1) Offline module (Indexing) – builds and maintains the index of web pages.
  (2) Online module (Query Processing and Retrieval) – processes user queries and returns ranked results.
- Main components:
  1. Web Crawler
  2. Page Repository
  3. Indexer
  4. Index (Inverted Index)
  5. Query Processor
  6. Ranking Module
  7. Results Presentation

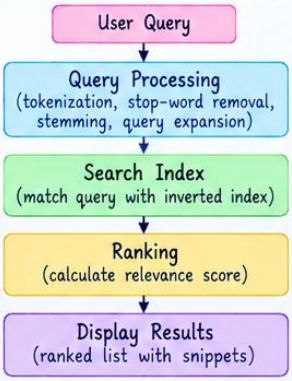
4) Search Engine Architecture Diagram :



5) Operation Stages :

(1) Crawling	• Web crawler (spider) downloads web pages from the Internet by following links. • Discovered pages are stored in a repository.
(2) Parsing	• Extracts useful information from pages (text, title, links, metadata). • Cleans the content by removing HTML tags, scripts, etc.
(3) Indexing	• Processed terms are stored in an inverted index. Each term is mapped to a list of documents with term frequency and position.
(4) Query Processing	• User query is tokenized, stop-words are removed, stemming/lemmatization may be applied. • Query can be expanded using synonyms or related terms.
(5) Searching & Ranking	• System matches query terms with index. • Documents are ranked using similarity measures (e.g., TF-IDF, BM25, PageRank, etc.).
(6) Result Presentation	• Top ranked results are shown to the user with title, URL and short snippet. • Results may include web pages, images, videos, news, etc.

6) Retrieval Process (for a query) :



7) Advantages of this Architecture :

- Efficient handling of large scale data.
- Fast search using inverted index.
- Modular design (separate offline and online processing).
- Supports different types of media (text, images, videos).
- Provides relevant and ranked results to users.

8) Conclusion :

Search engine architecture works in two phases – indexing (offline) and query processing (online) – to efficiently retrieve and rank relevant information from the web, fulfilling the purpose of providing fast, accurate and useful results to users.

Q9.

ANS

Q. Starting points, collection lists and overviews (10 Marks)

Explain browsing entry points, collection lists and collection overviews.

1) Introduction

- In Information Retrieval (IR), searching is not the only way to find information.
- Users can also browse a collection using different starting points, collection lists and overviews.
- These browsing methods help users explore the available content, understand the collection and select useful documents without formulating a query.
- They are especially useful for new users or when the user is not sure about the exact query terms.

2) Browsing Entry Points (Starting Points)

- Browsing entry points are initial access points from which a user can start exploring a collection.
- They provide different ways to navigate the collection based on document organization and metadata.
- They guide users to relevant parts of the collection even without a specific query.

Common Browsing Entry Points :

1. Alphabetical browsing (A-Z list of authors, titles, subjects)
2. Subject/category browsing (e.g., topics, domains)
3. Hierarchical browsing (from general to specific)
4. Temporal browsing (by year, date)
5. Geographical browsing (by location)
6. Faceted browsing (using multiple filters like author, type, year)

Example :



3) Collection Lists

- A collection list is a structured list of items that are available in a digital collection.
- It shows the documents or categories of documents present in the collection.
- Collection lists help users to see what is available and choose items for further browsing or retrieval.
- The list may contain titles, authors, subjects, dates or document types.

Example : Collection List (Document Titles)

No.	Document Title	Author	Year
1	Data Mining Concepts	Han	2021
2	Information Retrieval	Manning	2020
3	Machine Learning Basics	Mitchell	2019
4	Web Search Techniques	Baeza-Yates	2022
5	Natural Language Processing	Jurafsky	2021

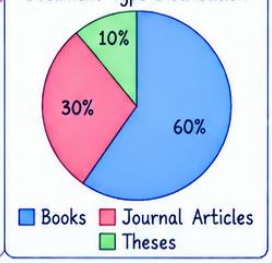
4) Collection Overviews

- A collection overview provides a summary of the entire collection to help users understand its content, scope and structure.
- It gives high-level information rather than listing individual documents.
- Overviews help users decide which part of the collection is relevant to their information need.

Example : Collection Overview

Digital Library Collection Overview

Total Documents	:	50,000
Subjects	:	Computer Science, Information Technology, Mathematics, Electronics
Document Types	:	Books (60%), Journal Articles (30%), Theses (10%)
Time Span	:	1990 – 2024
Languages	:	English (95%), Others (5%)



5) Advantages and Importance

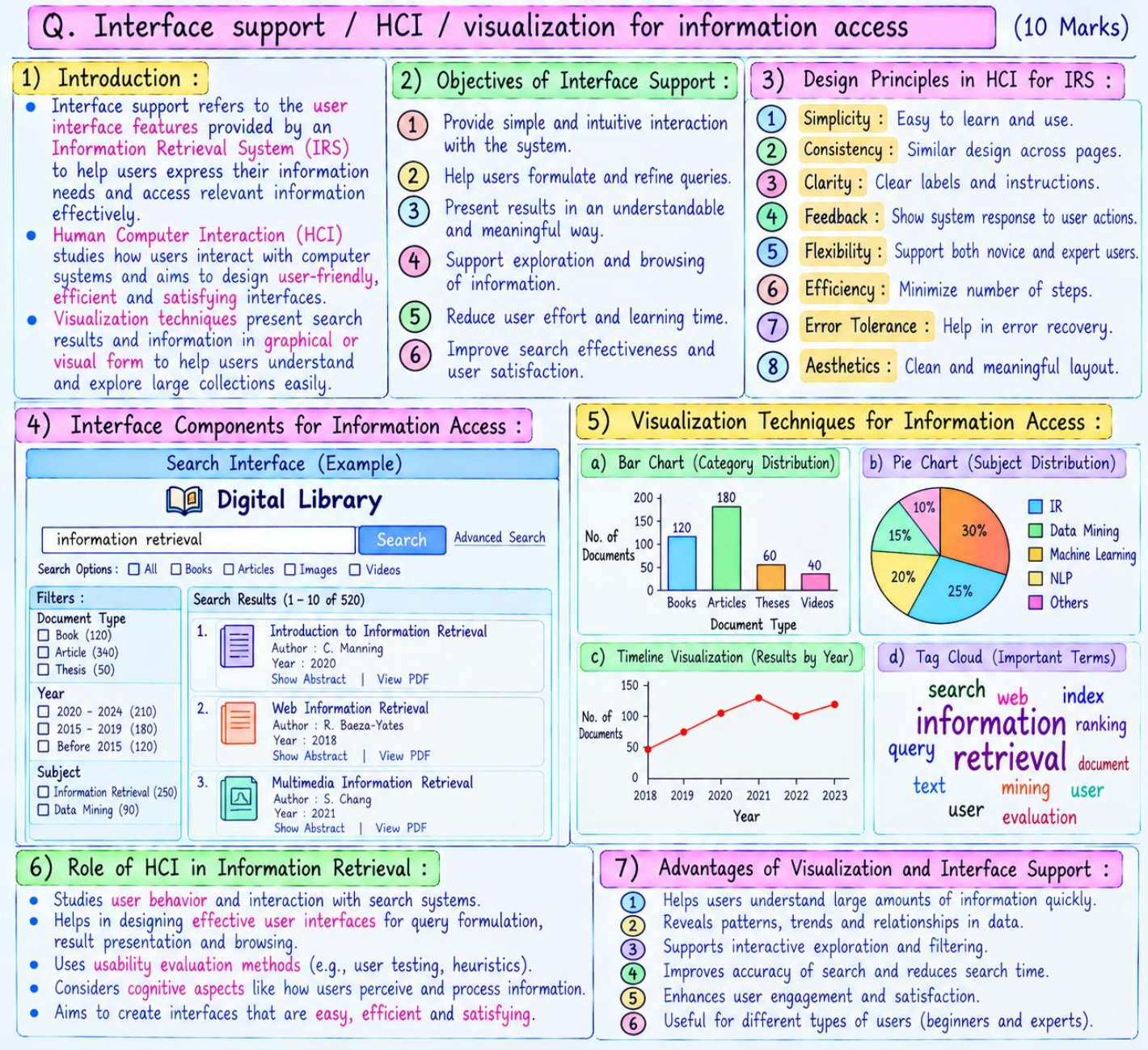
- Helps users explore the collection effectively.
- Useful when the user does not know exact query terms.
- Provides multiple pathways to relevant information.
- Improves user satisfaction and search experience.
- Supports both novice and expert users.

5) Conclusion

Browsing entry points, collection lists and collection overviews are important features in IR systems. They provide alternative ways to access and explore the collection, help users understand its content and structure, and assist in finding relevant documents even without a specific query.

Q10.

ANS


Q. Interface support / HCI / visualization for information access (10 Marks)
1) Introduction :
Interface support refers to the user interface features provided by an Information Retrieval System (IRS) to help users express their information needs and access relevant information effectively.
Human Computer Interaction (HCI) studies how users interact with computer systems and aims to design user-friendly, efficient and satisfying interfaces.
Visualization techniques present search results and information in graphical or visual form to help users understand and explore large collections easily.
2) Objectives of Interface Support :
1 Provide simple and intuitive interaction with the system.
2 Help users formulate and refine queries.
3 Present results in an understandable and meaningful way.
4 Support exploration and browsing of information.
5 Reduce user effort and learning time.
6 Improve search effectiveness and user satisfaction.
3) Design Principles in HCI for IRS :
1 Simplicity : Easy to learn and use.
2 Consistency : Similar design across pages.
3 Clarity : Clear labels and instructions.
4 Feedback : Show system response to user actions.
5 Flexibility : Support both novice and expert users.
6 Efficiency : Minimize number of steps.
7 Error Tolerance : Help in error recovery.
8 Aesthetics : Clean and meaningful layout.
4) Interface Components for Information Access :
Search Interface (Example)
Digital Library
information retrieval
Search
Advanced Search
Search Options : All Books Articles Images Videos
Filters :
Document Type
Book (120)
Article (340)
Thesis (50)
Year
2020 - 2024 (210)
2015 - 2019 (180)
Before 2015 (120)
Subject
Information Retrieval (250)
Data Mining (90)
Search Results (1 - 10 of 520)
1. Introduction to Information Retrieval
Author : C. Manning
Year : 2020
Show Abstract | View PDF
2. Web Information Retrieval
Author : R. Baeza-Yates
Year : 2018
Show Abstract | View PDF
3. Multimedia Information Retrieval
Author : S. Chang
Year : 2021
Show Abstract | View PDF
5) Visualization Techniques for Information Access :
a) Bar Chart (Category Distribution)
No. of Documents
200
150
100
50
0
120
180
60
40
Books
Articles
Theses
Videos
Document Type
b) Pie Chart (Subject Distribution)
10%
30%
15%
20%
25%
IR
Data Mining
Machine Learning
NLP
Others
c) Timeline Visualization (Results by Year)
No. of Documents
150
100
50
0
2018
2019
2020
2021
2022
2023
Year
d) Tag Cloud (Important Terms)
search
web
index
information
ranking
query
retrieval
document
text
mining
user
user
evaluation
6) Role of HCI in Information Retrieval :
Studies user behavior and interaction with search systems.
Helps in designing effective user interfaces for query formulation, result presentation and browsing.
Uses usability evaluation methods (e.g., user testing, heuristics).
Considers cognitive aspects like how users perceive and process information.
Aims to create interfaces that are easy, efficient and satisfying.
7) Advantages of Visualization and Interface Support :
1 Helps users understand large amounts of information quickly.
2 Reveals patterns, trends and relationships in data.
3 Supports interactive exploration and filtering.
4 Improves accuracy of search and reduces search time.
5 Enhances user engagement and satisfaction.
6 Useful for different types of users (beginners and experts).

Q11.

ANS

Q. Markup languages and applications. Prepare any two languages and applications. (10 Marks)

1) HTML (HyperText Markup Language)

1.1 Introduction :

- HTML is a standard markup language used to create and structure web pages on the World Wide Web.
- It uses predefined tags to define the structure and content of a web document.
- It is the backbone of web development and is interpreted by web browsers.

1.2 Basic Structure and Example :

HTML Code :

```
<!DOCTYPE html>
<html>
  <head>
    <title>Information Retrieval</title>
  </head>
  <body>
    <h1>Welcome to IRS</h1>
    <p>Information Retrieval deals with
     searching and accessing information.</p>
  </body>
</html>
```

Output in Browser :



1.3 Important HTML Tags :

Tag	Purpose
<html> ... </html>	Defines the HTML document
<head> ... </head>	Contains meta-information
<title> ... </title>	Specifies the title of the page
<body> ... </body>	Contains visible page content
...	Defines a heading
...	Defines a paragraph
<a> ... 	Creates a hyperlink
<img>	Inserts an image

1.4 Applications of HTML :

1. Creating web pages and web applications.
2. Used in digital libraries to present search results.
3. Used to design user interfaces for IR systems.
4. Supports hyperlinks for easy navigation.
5. Works with CSS and JavaScript to create interactive pages.

2) XML (eXtensible Markup Language)

2.1 Introduction :

- XML is a flexible markup language used to store, represent and transport structured data.
- It allows users to define their own tags (unlike HTML).
- It is platform independent, self-describing and widely used for data exchange between applications.

2.2 Basic Structure and Example :

XML Code :

```
<?xml version="1.0"?>
<book>
  <title>Information Retrieval</title>
  <author>C. Manning</author>
  <year>2020</year>
  <publisher>Cambridge</publisher>
</book>
```

Tree Structure :



2.3 Features of XML :

- User-defined tags.
- Hierarchical structures.
- Separation of data and presentation.
- Extensible and platform independent.
- Supports metadata and complex data.
- Used in many IR and web services applications.

2.4 Applications of XML :

1. Data exchange between different systems (e.g., search engines).
2. Representing and storing metadata.
3. Used in digital libraries (e.g., metadata in Dublin Core).
4. Configuration files (e.g., for IR tools).
5. Web services and APIs (e.g., RSS, XML for search results).

3) Comparison between HTML and XML :

Aspect	HTML	XML
Purpose	Used to display web pages.	Used to store and transport structured data.
Tags	Predefined tags (e.g., <html>,).	User-defined tags (e.g., <book>, <author>).
Focus	Presentation of data.	Description and storage of data.
Case Sensitivity	Not case sensitive.	Case sensitive.
Structure	Relatively simple.	Strictly hierarchical.
Application	Web page design, user interfaces.	Data exchange, metadata, configuration, web services.

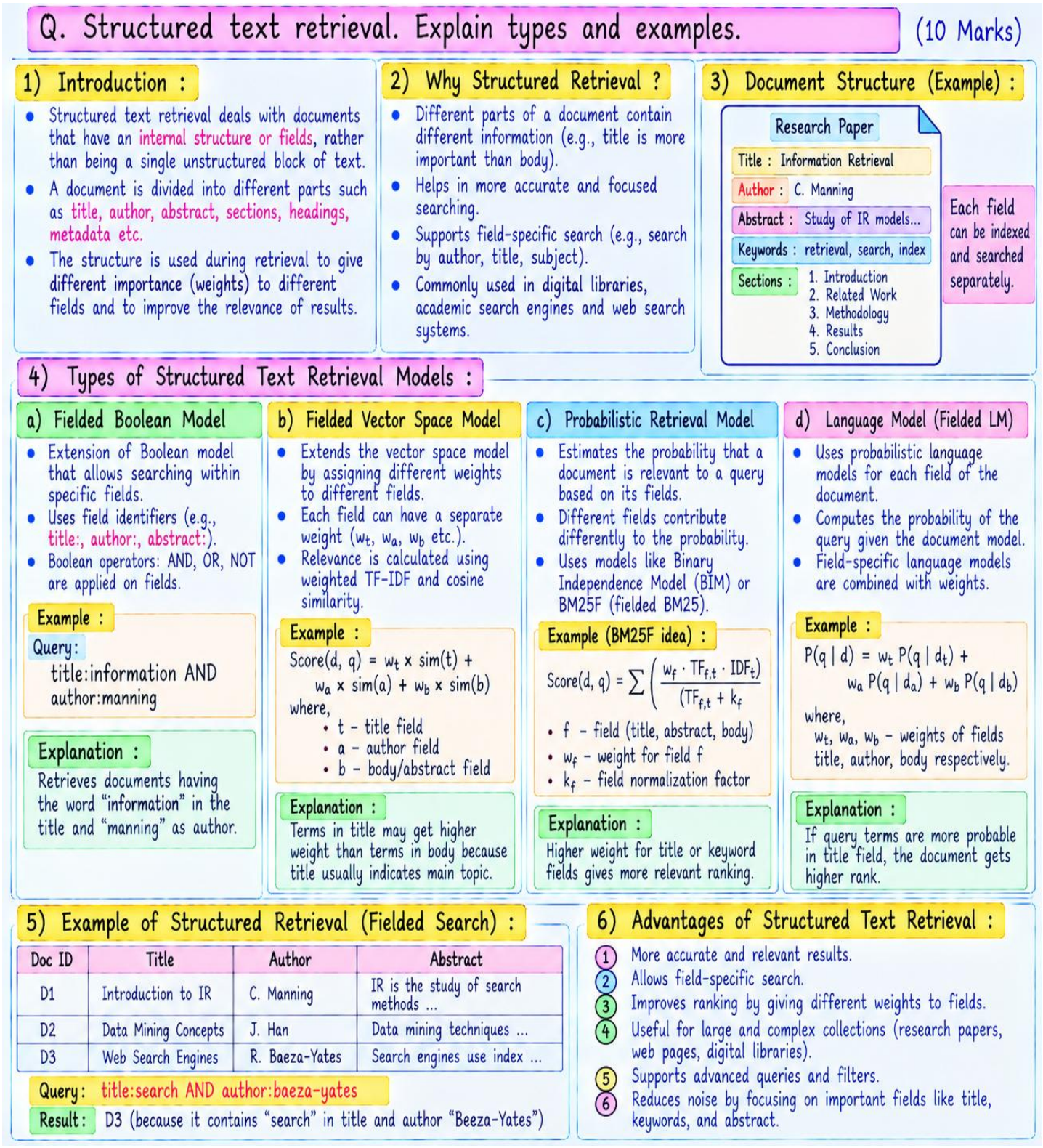
Q12.

ANS


Q. Taxonomy / classification of IR models and browsing models. Draw classification tree and explain major model families. (10 Marks)
Information Retrieval (IR) Models
IR Models (Query-based retrieval models)
Browsing Models (Exploration-based models)
Boolean Models
Vector Space Models
Probabilistic Models
Language Models
Neural / Learning-based Models
Hierarchical Browsing
Faceted Browsing
Attribute-based Browsing
• Boolean Algebra • Exact match retrieval • Uses AND, OR, NOT operators
• Represents documents and queries as vectors • Uses term weights (TF-IDF) • Ranks documents using similarity (e.g., cosine similarity)
• Based on probability theory • Estimates relevance of documents • Examples: Binary Independence Model, BM25
• Uses probabilistic language modelling • Estimates probability of query given a document (P(Q|D)) • Examples: Query likelihood model, Dirichlet smoothing
• Uses machine learning / deep learning • Learn semantic representations • Examples: Word2Vec, BERT, Transformers
• Uses a hierarchy (taxonomy) of topics/categories • User navigates from general to specific • Example: Subject directories (e.g., Yahoo! Directory)
• Uses multiple independent facets/attributes • Users can combine filters • Example: Filter by author, year, document type, subject
• Uses metadata attributes for browsing • Example attributes: Author, Title, Year, Location, Document type
1) Major IR Model Families
1 Boolean Model
• Documents and queries are represented using Boolean algebra.
• Uses logical operators: AND, OR, NOT.
• Retrieves documents that exactly match the query condition.
• Suitable for simple keyword queries.
• Example: Information about "machine AND learning".
• Advantage: Simple and easy to implement.
• Limitation: Does not consider partial matching or ranking.
2 Vector Space Model (VSM)
• Represents each document and query as a vector of term weights.
• Uses weighting schemes like TF-IDF.
• Ranks documents based on similarity (e.g., cosine similarity).
• Advantage: Handles partial matching and ranking.
• Limitation: Ignores term dependencies and semantics.
3 Probabilistic Model
• Based on probability theory and decision theory.
• Estimates the probability of a document being relevant to a query.
• Examples: Binary Independence Model, BM25.
• Advantage: Provides a principled probabilistic framework.
• Limitation: Requires training data or parameter estimation.
4 Language Model
• Treats each document as a language model.
• Estimates the probability of generating the query from a document (P(Q|D)).
• Uses smoothing techniques (e.g., Dirichlet, Jelinek-Mercer).
• Advantage: Effective for handling term variations.
• Limitation: Computationally more complex.
5 Neural / Learning-based Model
• Uses machine learning and deep learning to learn semantic representations of text.
• Examples: Word2Vec, Doc2Vec, BERT, Transformers.
• Can capture context and meaning beyond exact keywords.
• Advantage: High accuracy, handles synonymy and context.
• Limitation: Requires large training data and more computational resources.
2) Major Browsing Model Families
1 Hierarchical Browsing Model
• Organizes documents in a hierarchy of categories (taxonomy).
• User navigates from general categories to specific subcategories.
• Example: Yahoo! Directory, Library classification.
• Advantage: Easy to understand, suitable for exploratory search.
• Limitation: Rigid hierarchy, may not fit all documents.
2 Faceted Browsing Model
• Uses multiple independent facets (e.g., author, subject, year, type).
• Users can combine different facet values to narrow down results.
• Example: Digital libraries, e-commerce websites.
• Advantage: Flexible, supports multi-criteria browsing.
• Limitation: Requires good metadata and facet design.
3 Attribute-based Browsing Model
• Browses the collection using metadata attributes.
• Common attributes: Author, Title, Subject, Year, Document type, Location.
• Example: Library catalogues, institutional repositories.
• Advantage: Simple and effective for well-structured collections.
• Limitation: Depends on quality of metadata.
3) Comparison (IR Models vs Browsing Models)
Aspect | IR Models (Query-based) | Browsing Models (Exploration-based)
Approach | Uses a query to retrieve documents | Navigates through categories/attributes
User intent | Known information need | Exploratory or uncertain need
Input | Text query | Browsing selection (category, facet, etc.)
Result | Ranked list of relevant documents | List of documents in selected category/facet
Examples | Boolean, VSM, Probabilistic, Language, Neural models | Hierarchical, Faceted, Attribute-based
4) Conclusion
• IR models and browsing models are complementary approaches.
• IR models are useful when the user has a specific query, while browsing models help users explore and discover relevant information.
• A good IR system often combines both approaches for better search experience.

Q13.

ANS


Q. Structured text retrieval. Explain types and examples. (10 Marks)
1) Introduction :
Structured text retrieval deals with documents that have an internal structure or fields, rather than being a single unstructured block of text.
A document is divided into different parts such as title, author, abstract, sections, headings, metadata etc.
The structure is used during retrieval to give different importance (weights) to different fields and to improve the relevance of results.
2) Why Structured Retrieval ?
Different parts of a document contain different information (e.g., title is more important than body).
Helps in more accurate and focused searching.
Supports field-specific search (e.g., search by author, title, subject).
Commonly used in digital libraries, academic search engines and web search systems.
3) Document Structure (Example) :
Research Paper
Title : Information Retrieval
Author : C. Manning
Abstract : Study of IR models...
Keywords : retrieval, search, index
Sections : 1. Introduction 2. Related Work 3. Methodology 4. Results 5. Conclusion
Each field can be indexed and searched separately.
4) Types of Structured Text Retrieval Models :
a) Fielded Boolean Model
Extension of Boolean model that allows searching within specific fields.
Uses field identifiers (e.g., title:, author:, abstract:).
Boolean operators: AND, OR, NOT are applied on fields.
Example :
Query: title:information AND author:manning
Explanation :
Retrieves documents having the word "information" in the title and "manning" as author.
b) Fielded Vector Space Model
Extends the vector space model by assigning different weights to different fields.
Each field can have a separate weight (w_t, w_a, w_b etc.).
Relevance is calculated using weighted TF-IDF and cosine similarity.
Example :
$Score(d, q) = w_t \times sim(t) + w_a \times sim(a) + w_b \times sim(b)$
where,
t - title field
a - author field
b - body/abstract field
Explanation :
Terms in title may get higher weight than terms in body because title usually indicates main topic.
c) Probabilistic Retrieval Model
Estimates the probability that a document is relevant to a query based on its fields.
Different fields contribute differently to the probability.
Uses models like Binary Independence Model (BIM) or BM25F (fielded BM25).
Example (BM25F idea) :
$Score(d, q) = \sum \left(\frac{w_f \cdot TF_{f,t} \cdot IDF_t}{(TF_{f,t} + k_f} \right)$
f - field (title, abstract, body)
w_f - weight for field f
k_f - field normalization factor
Explanation :
Higher weight for title or keyword fields gives more relevant ranking.
d) Language Model (Fielded LM)
Uses probabilistic language models for each field of the document.
Computes the probability of the query given the document model.
Field-specific language models are combined with weights.
Example :
$P(q \mid d) = w_t P(q \mid d_t) + w_a P(q \mid d_a) + w_b P(q \mid d_b)$
where,
w_t, w_a, w_b - weights of fields title, author, body respectively.
Explanation :
If query terms are more probable in title field, the document gets higher rank.
5) Example of Structured Retrieval (Fielded Search) :
Doc ID | Title | Author | Abstract
D1 | Introduction to IR | C. Manning | IR is the study of search methods ...
D2 | Data Mining Concepts | J. Han | Data mining techniques ...
D3 | Web Search Engines | R. Baeza-Yates | Search engines use index ...
Query: title:search AND author:baeza-yates
Result: D3 (because it contains "search" in title and author "Beeza-Yates")
6) Advantages of Structured Text Retrieval :
1 More accurate and relevant results.
2 Allows field-specific search.
3 Improves ranking by giving different weights to fields.
4 Useful for large and complex collections (research papers, web pages, digital libraries).
5 Supports advanced queries and filters.
6 Reduces noise by focusing on important fields like title, keywords, and abstract.

UT-2

Q1. .Discuss various phases of text preprocessing within a document. Discuss any one application for the same.

ANS

Text Preprocessing in Information Retrieval System

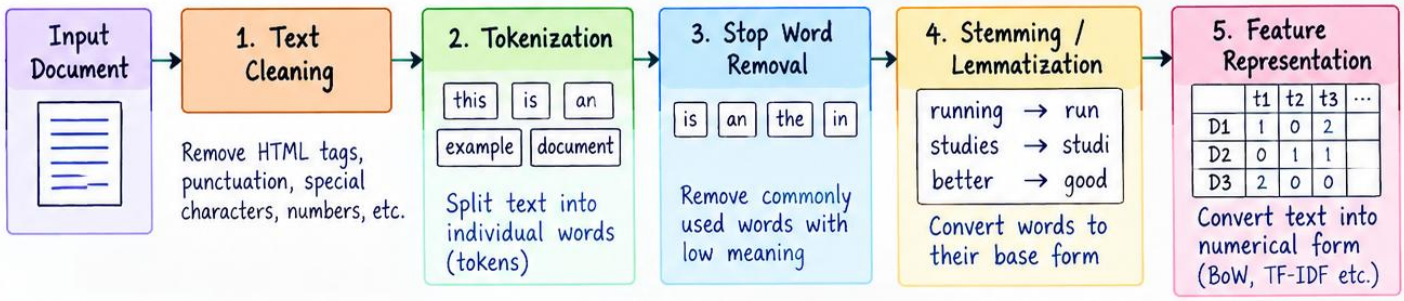
1. Introduction

Text preprocessing is the first and most important step in an Information Retrieval (IR) system. It converts raw, unstructured text documents into a clean and normalized form so that they can be easily indexed and retrieved. The main goal is to remove noise, reduce dimensionality, and improve the quality of representation of text.

Objectives of Text Preprocessing

- Remove unnecessary data (noise)
- Reduce the size of vocabulary
- Improve the effectiveness of retrieval
- Normalize different forms of words
- Make the text suitable for indexing and analysis

2. Phases of Text Preprocessing



1. Text Cleaning (Noise Removal)

- Removes unwanted elements such as HTML tags, URLs, punctuation marks, special characters, numbers, and extra spaces.
- Converts all text to lowercase to maintain uniformity.

Example:

Input : This is an Example!!! 123
After Cleaning : this is an example

2. Tokenization

- Splits the cleaned text into individual words or tokens.
- Tokens are the basic units for further processing.

Example:

Input : this is an example
Tokens : [this] [is] [an] [example]

3. Stop Word Removal

- Removes very common words (e.g., is, an, the, in, at, to) which do not carry significant meaning for retrieval.
- Helps in reducing the size of the index.

Example:

Input tokens : [this] [is] [an] [example]
After removal : [this] [example]

4. Stemming / Lemmatization

- Converts words to their root or base form to treat different forms of a word as one.
- Stemming: removes suffixes (e.g., running → run).
- Lemmatization: converts to meaningful base form using a dictionary (e.g., better → good).

Example:

Input : [running] [studies] [documents]
After Stemming : [run] [studi] [document]

5. Feature Representation

- Converts preprocessed text into numerical form for indexing and retrieval.
- Common methods:
  - Bag of Words (BoW)
  - Term Frequency – Inverse Document Frequency (TF-IDF)
  - Word Embeddings (e.g., Word2Vec, BERT)

3. Application : Document Search / Web Search

- Text preprocessing is widely used in search engines (e.g., Google) to improve retrieval of relevant documents.
- It helps in removing noise, handling different word forms, and creating an efficient index.
- When a user enters a query, the same preprocessing steps are applied to the query. Then the system retrieves documents that are most relevant to the processed query using similarity measures (e.g., TF-IDF, cosine similarity).

Q2. Write a note on Scalar and Metrics Clusters.

ANS

Scalar and Metric Clusters (Clustering)

1. Introduction

- Clustering is an unsupervised learning technique used to group similar objects or documents together.
- In Information Retrieval (IR), clustering helps to organize a large collection of documents so that similar documents fall into the same group.
- Clusters can be formed based on the type of data representation and the similarity measure used.
- Two commonly discussed types are Scalar Clusters and Metric Clusters.

2. Scalar Clusters

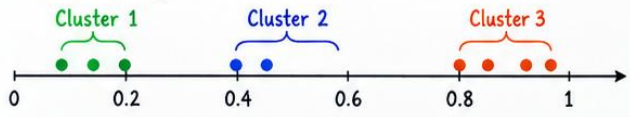
- In scalar clustering, each object or document is represented using a single numerical value (scalar).
- Clustering is done based on similarity or distance between these scalar values, usually on a 1-dimensional scale.
- Objects with close scalar values are grouped into the same cluster.
- It is simple and mainly used when data has only one feature or can be reduced to a single numeric score (e.g., document relevance score, term frequency, rating etc.).

Example : Scalar Clustering

Consider the following document relevance scores:

Document	D1	D2	D3	D4	D5	D6	D7	D8
Score	0.1	0.2	0.15	0.8	0.85	0.78	0.4	0.45

After clustering:



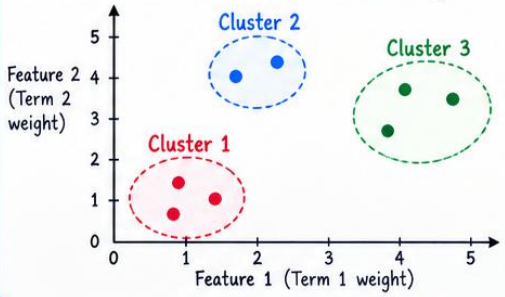
Documents having similar scalar values are grouped together.

3. Metric Clusters

- In metric clustering, each object or document is represented by a multi-dimensional feature vector (set of attributes), and clustering is done using a distance or similarity measure (called a metric).
- It considers multiple features of the data, such as term weights, TF-IDF values, or semantic features.
- Common distance/metric measures include:
  - Euclidean distance
  - Manhattan distance
  - Cosine similarity (widely used in IR for text data)
- Objects (documents) that are close to each other in the multi-dimensional space are grouped into the same cluster.
- It is more general and widelly used in Information Retrieval because documents are usually represented by many features.

Example : Metric Clustering (2-D representation)

Consider documents represented using two features (Feature 1 and Feature 2) such as term weights.



Document	(F1, F2)
D1	(1, 1)
D2	(1.5, 1)
D3	(1, 1.5)
D4	(4, 4)
D5	(4.5, 4)
D6	(4, 3.5)
D7	(2, 4)
D8	(2.5, 4.5)

4. Comparison

Aspect	Scalar Clusters	Metric Clusters
Data representation	Single numerical value (1-D)	Multi-dimensional feature vector
Similarity measure	Based on scalar distance (e.g., $\|x - y\|$)	Based on distance/metric (e.g., Euclidean, Cosine)
Complexity	Simple	More complex
Use in IR	Used when only one feature is considered (e.g., relevance score)	Widely used for document clustering in IR
Example	Relevance scores, ratings	TF-IDF vectors, term weights

Introduction

Clustering is an unsupervised learning technique used to group similar objects or documents into the same cluster. In an Information Retrieval (IR) system, clustering helps organize a large collection of documents so that similar documents can be easily searched and retrieved.

Scalar and metric clusters are two approaches to grouping data based on its representation and the similarity or distance between objects.

2. Scalar Clusters

A scalar cluster is a group of objects or documents represented using a single numerical value called a scalar. Objects having similar or nearby scalar values are grouped together.

Characteristics

1. Each object is represented by a single numerical value.
2. Clustering is performed using the difference between scalar values.
3. It is simple and easy to understand.
4. It is useful when data can be represented by one numerical feature or score.

Example: In an Information Retrieval System, documents can be grouped according to their relevance scores.

Suppose the relevance scores are:

Document	Relevance score
D1	0.10
D2	0.15
D3	0.20
D4	0.75
D5	0.80
D6	0.85

The documents can be grouped into two clusters:

- Cluster 1: D1, D2, D3
- Cluster 2: D4, D5, D6

Documents with similar scores are placed in the same cluster.

3. Metric Clusters

A metric cluster is a group of objects or documents represented using multiple features. The similarity between objects is determined using a distance or similarity measure, such as Euclidean distance or cosine similarity.

Characteristics

1. Each object is represented by a feature vector containing multiple values.
2. Clustering is based on the distance or similarity between feature vectors.
3. It can consider multiple characteristics of a document.
4. It is useful for complex datasets and document collections.

Example: In an Information Retrieval System, a document may be represented using the weights of different terms.

Suppose documents are represented by two features:

- D1 = (1, 2)
- D2 = (1.2, 2.1)
- D3 = (8, 9)
- D4 = (8.2, 9.1)

The documents can be grouped as:

- Cluster 1: D1, D2
- Cluster 2: D3, D4

D1 and D2 are grouped together because their feature values are close. Similarly, D3 and D4 form another cluster.

Common distance and similarity measures

- Euclidean distance: Measures the straight-line distance between two points.
- Manhattan distance: Measures the sum of absolute differences between feature values.
- Cosine similarity: Measures the similarity between two vectors based on the angle between them. It is commonly used for text documents.

4. Difference Between Scalar and Metric Clusters

Basis	Scalar Clusters	Metric Clusters
Representation	Single numerical value	Multiple numerical features
Dimensions	Usually one-dimensional	Often multidimensional
Comparison	Difference between scalar values	Distance or similarity between vectors
Complexity	Simple	Comparatively complex
Example	Document relevance scores	Document term-weight vectors
Application	Grouping by one score	Grouping similar documents by multiple features

5. Applications in Information Retrieval

Scalar and metric clustering can be used in the following applications:

1. Document organization: Groups similar documents together.
2. Search engines: Helps organize documents and improve access to relevant information.
3. Search result analysis: Scalar scores can help group documents according to relevance.
4. Document recommendation: Metric clustering can identify documents similar to those a user has selected.
5. Information filtering: Helps categorize large document collections into meaningful groups.

6. Advantages

- Organizes large amounts of information.
- Helps identify similarities among documents.
- Makes document analysis easier.
- Supports efficient searching and information retrieval.
- Metric clustering can handle documents represented by multiple features.

Q3. What is flat browsing and hypertext browsing?

ANS

Flat Browsing and Hypertext Browsing

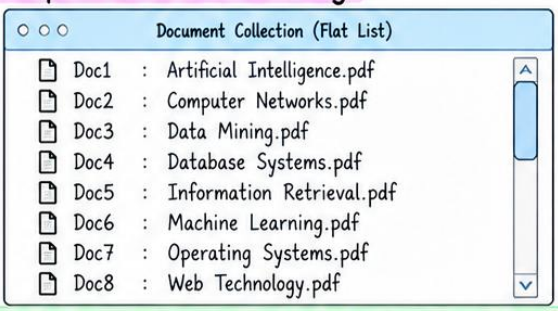
1. Introduction

- Browsing is a method of searching and exploring information without writing complex queries.
- It allows users to navigate through a collection of documents or web pages and find relevant information.
- In Information Retrieval (IR) systems, browsing can be done in two main ways:
  1) Flat browsing
  2) Hypertext browsing

2. Flat Browsing

- Flat browsing is a simple form of browsing where documents are organized in a single-level list or alphabetical order.
- All documents are shown at the same level without hierarchical structure.
- Users can scroll through the list and select the document that seems relevant.
- It is similar to browsing a list of files in a folder.
- It is easy to implement and understand, but it can become difficult when the collection is very large.

3. Example of Flat Browsing

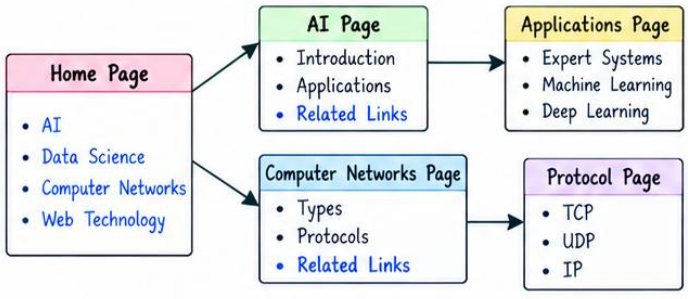


All documents are shown in a single list at the same level. User can scroll and select any document.

4. Hypertext Browsing

- Hypertext browsing is a more advanced form of browsing where documents are connected through hyperlinks.
- Documents are organized in a network or web structure.
- Users can move from one document to another by clicking on links.
- It allows non-linear navigation, meaning users can follow different paths based on their interest.
- It is widely used on the World Wide Web (WWW).
- It helps users to explore related information easily.

5. Example of Hypertext Browsing



Documents are connected through hyperlinks, allowing users to navigate from one page to another.

6. Comparison between Flat and Hypertext Browsing

Aspect	Flat Browsing	Hypertext Browsing
Structure	Single list, no hierarchy	Network of linked documents
Navigation	Linear (one by one)	Non-linear (via hyperlinks)
Organization	All documents at same level	Documents connected by links
User control	Limited	More flexible
Ease of use	Simple and easy	More interactive
Suitable for	Small or medium collections	Large and complex collections (e.g., Web)
Example	List of PDF files in a folder	Web pages linked through URLs

7. Advantages and Disadvantages

Flat Browsing

- **Advantages:** Simple, easy to implement, suitable for small collections.
- **Disadvantages:** Not suitable for large collections, no relation between documents.

Hypertext Browsing

- **Advantages:** Easy navigation, access to related information, suitable for large collections.
- **Disadvantages:** Can be complex, users may get lost (information overload).

Introduction

Browsing is a method of searching and exploring information without necessarily writing a complex search query. It allows users to navigate through a collection of documents or web pages to find the required information.

In an Information Retrieval System, two common browsing methods are:

1. Flat Browsing
2. Hypertext Browsing

2. Flat Browsing

Definition: Flat browsing is a browsing technique in which documents or information items are displayed in a single-level list without links connecting them in a hierarchical or network structure.

Characteristics

1. Documents are displayed in a list or table.
2. All items are generally shown at the same level.
3. Users scroll through the list and select the required document.
4. It is simple to design and easy to understand.
5. It is suitable for small or medium-sized collections of information.

Example

Suppose a folder contains the following PDF files:

- Artificial Intelligence.pdf
- Computer Networks.pdf
- Database Systems.pdf
- Information Retrieval.pdf
- Machine Learning.pdf

The user scrolls through the list and opens the required file.

Advantages

- Simple and easy to use.
- Easy to implement.

- Suitable for displaying a list of documents.
- Requires less complex navigation.

Disadvantages

- Finding information can be difficult when the list is very large.
- Related documents are not necessarily connected.
- Users may need to scroll through many items.

3. Hypertext Browsing

Definition: Hypertext browsing is a browsing technique in which documents or web pages are connected through hyperlinks. Users can click these links to move from one document to another and explore related information.

Characteristics

1. Documents are connected through hyperlinks.
2. Users can navigate between related pages.
3. It supports non-linear navigation, meaning users can choose different paths.
4. It helps users explore related information easily.
5. It is widely used on the World Wide Web (WWW).

Example

Suppose a user opens an Artificial Intelligence webpage. It contains links to:

- Machine Learning
- Deep Learning
- Natural Language Processing
- Expert Systems

By clicking any link, the user can open the related page and continue exploring other topics.

Advantages

- Provides quick access to related information.
- Supports flexible and non-linear navigation.

- Makes exploring large collections of documents easier.
- Helps users discover additional relevant information.

Disadvantages

- Too many links can confuse users.
- Users may lose track of their original page.
- Creating and maintaining links can be complex.

4. Difference Between Flat Browsing and Hypertext Browsing

Basis	**Flat Browsing**	**Hypertext Browsing**
Structure	Single-level list	Network of linked documents
Navigation	Mainly through the list	Through hyperlinks
Movement	Usually item by item	Non-linear movement
Relationship	Documents are not necessarily connected	Documents are connected by links
Complexity	Simple	Comparatively complex
User flexibility	Limited	High
Example	List of PDF files in a folder	Web pages connected through links

5. Applications in Information Retrieval Systems

Flat browsing:

- Browsing files in a folder.
- Viewing a list of library books.
- Displaying search results in a simple list.

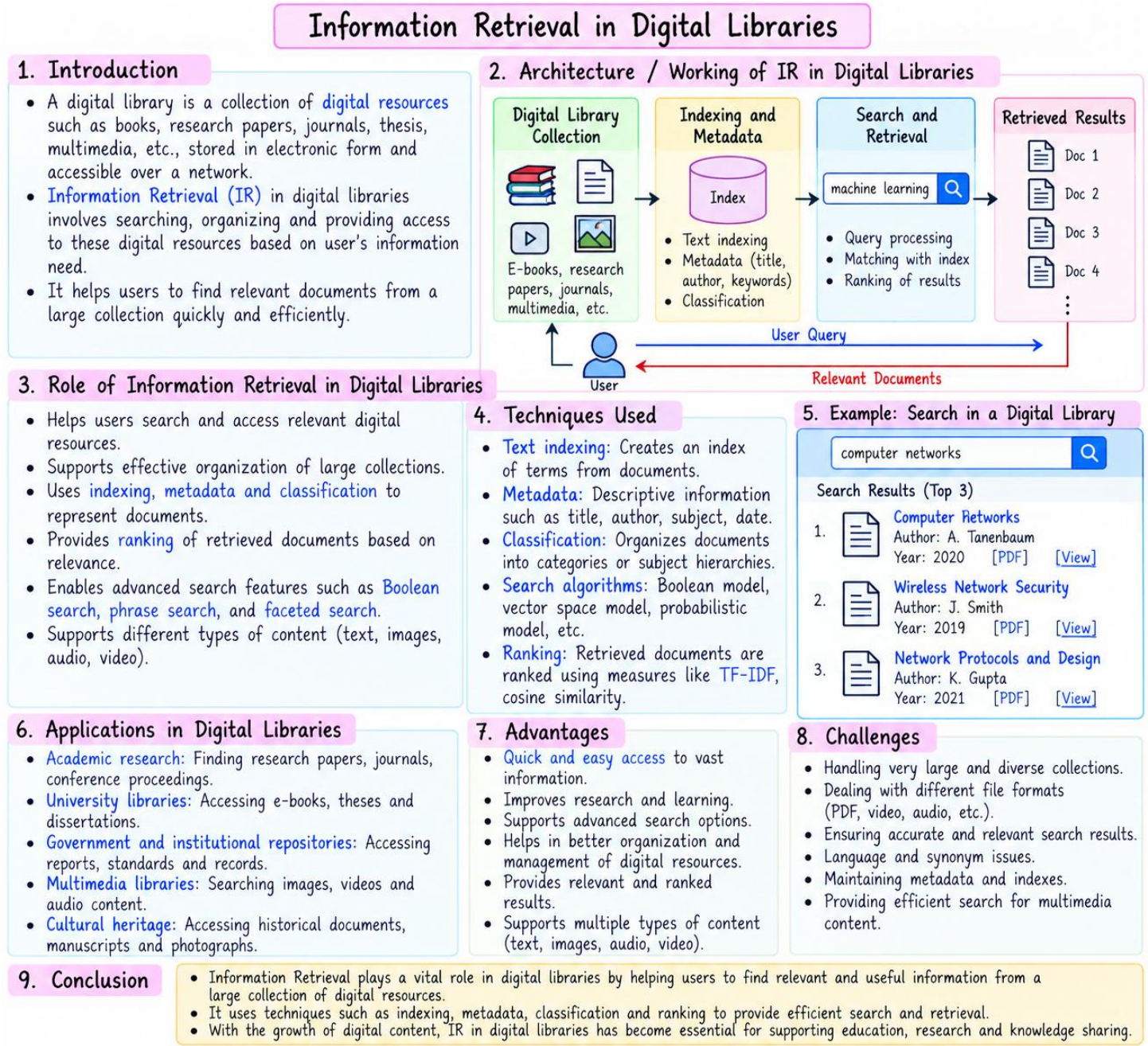
Hypertext browsing:

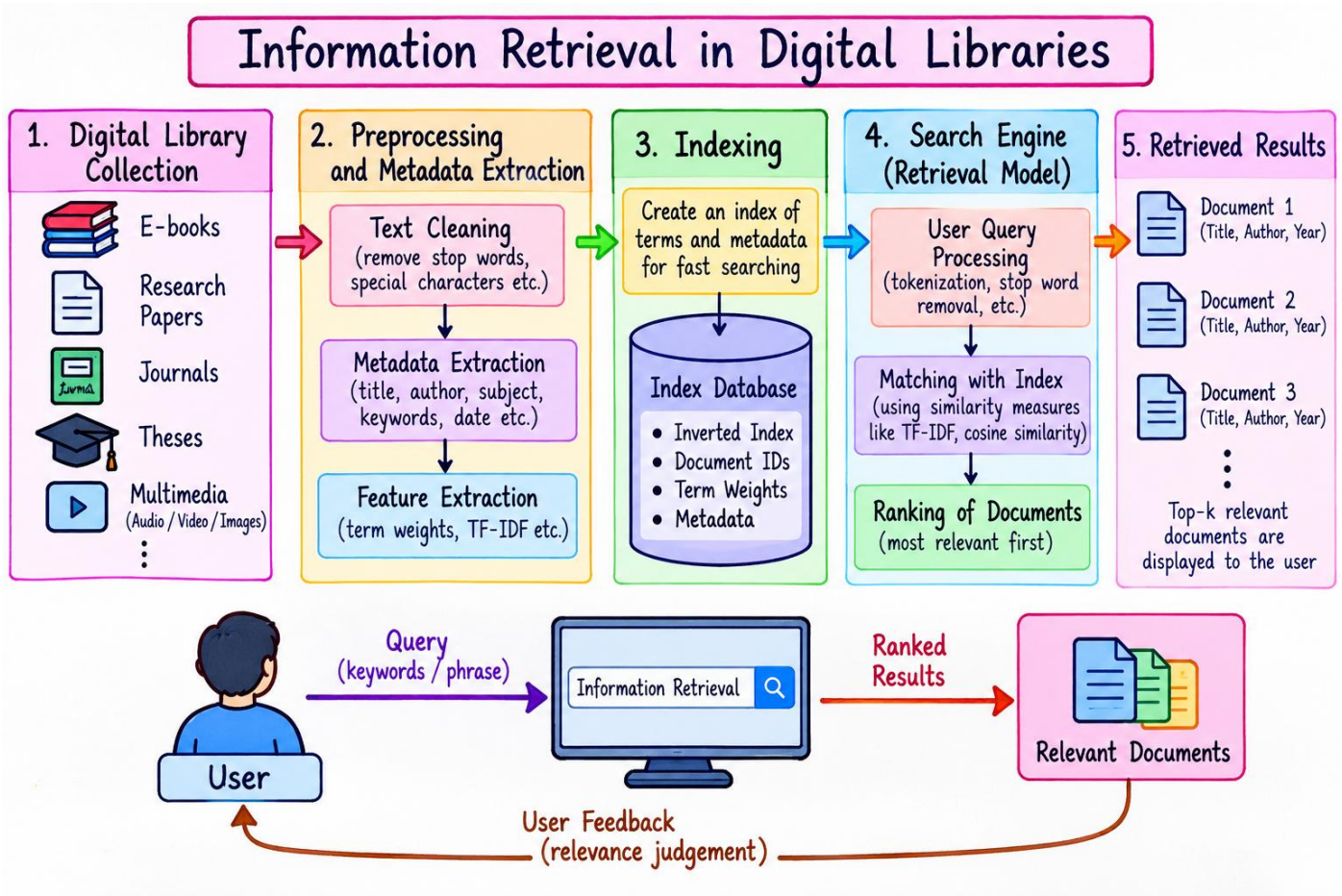
- Exploring websites and web pages.
- Navigating online documentation.

- Following references and links in digital libraries.
- Exploring related topics in online learning systems.

Q4. Write a Short Note on Information Retrieval in digital libraries.

ANS




Information Retrieval in Digital Libraries
1. Digital Library Collection
E-books
Research Papers
Journals
Theses
Multimedia
(Audio / Video / Images)
2. Preprocessing and Metadata Extraction
Text Cleaning
(remove stop words, special characters etc.)
Metadata Extraction
(title, author, subject, keywords, date etc.)
Feature Extraction
(term weights, TF-IDF etc.)
3. Indexing
Create an index of terms and metadata for fast searching
Index Database
Inverted Index
Document IDs
Term Weights
Metadata
4. Search Engine (Retrieval Model)
User Query Processing
(tokenization, stop word removal, etc.)
Matching with Index
(using similarity measures like TF-IDF, cosine similarity)
Ranking of Documents
(most relevant first)
5. Retrieved Results
Document 1
(Title, Author, Year)
Document 2
(Title, Author, Year)
Document 3
(Title, Author, Year)
Top-k relevant documents are displayed to the user
User
Query
(keywords / phrase)
Information Retrieval
Ranked Results
Relevant Documents
User Feedback
(relevance judgement)

Q5. Write a short note on Interface support for the search process.

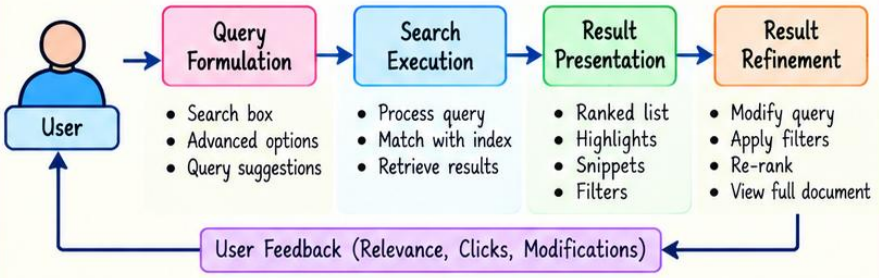
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Interface Support for the Search Process

1. Introduction

- The user interface acts as a **bridge** between the user and the information retrieval (IR) system.
- Interface support for the search process provides **tools and features** that help users express their information need, perform the search, view and refine the results, and access relevant documents easily.
- A good interface makes the search process **simple, efficient and user-friendly.**

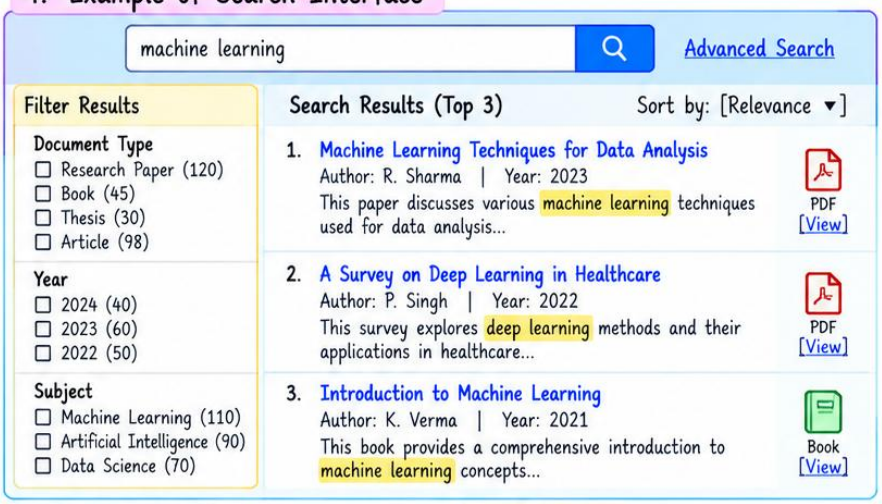
2. Role of Interface in the Search Process



3. Interface Features for Search Support

1. **Simple Search Box** – Allows users to enter keywords or phrases easily.
2. **Advanced Search Options** – Supports Boolean operators (AND, OR, NOT), phrase search, field search (title, author, date) etc.
3. **Query Suggestions** – Provides suggested terms automatically as the user types.
4. **Auto-complete** – Helps in faster and accurate query input.
5. **Filters and Facets** – Allows users to refine results by categories such as document type, author, year, subject etc.
6. **Result Presentation** – Shows a ranked list with titles, short snippets and highlighted keywords.
7. **Sorting Options** – Sort by relevance, date, author etc.
8. **Preview and Full Text Access** – Allows users to view a snippet or the full document.
9. **Query Refinement** – Enables users to modify the query, remove or add terms, or apply filters.
10. **Relevance Feedback** – Allows users to mark documents as relevant or non-relevant to improve future results.

4. Example of Search Interface



5. Advantages

- Makes search process easy and user-friendly.
- Helps users find relevant information quickly.
- Provides multiple ways to search and refine results.
- Supports efficient exploration of large collections.
- Improves user satisfaction and search accuracy.
- Saves time and effort.

6. Challenges

- Designing intuitive and easy-to-use interfaces.
- Handling complex queries and large result sets.
- Providing effective query suggestions and filters.
- Avoiding information overload.
- Supporting different types of users (beginners and experts).
- Keeping the interface simple yet powerful.

7. Conclusion

Interface support plays a vital role in the search process of an information retrieval system. It provides various features such as search box, advanced search, query suggestions, filters, result presentation and query refinement, which help users to express their information need and find relevant documents efficiently.

Q6. What is human-computer interaction? List and discuss any four design principles for information access interfaces.

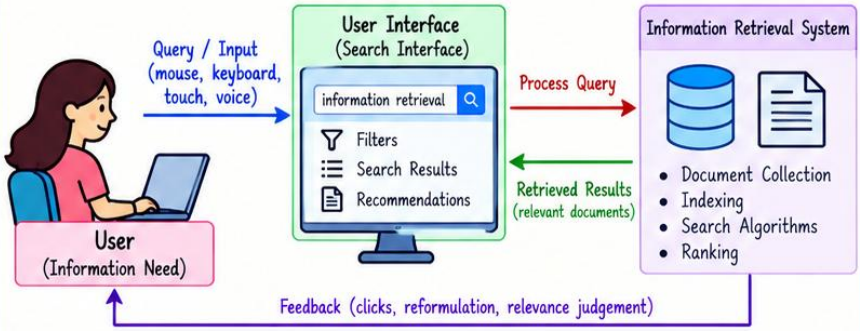
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Human-Computer Interaction and Design Principles for Information Access Interfaces

1. What is Human-Computer Interaction (HCI)?

- Human-Computer Interaction (HCI) is the study of how people interact with computers, and how to design computer systems that are easy, effective and pleasant to use.
- It involves understanding user needs, designing user-friendly interfaces, and evaluating how well users can achieve their goals.
- In the context of Information Retrieval (IR), HCI focuses on designing search interfaces that help users express their information needs and access relevant information easily and efficiently.

2. Human-Computer Interaction Model



3. Design Principles for Information Access Interfaces (Any Four)

1. Simplicity and Clarity

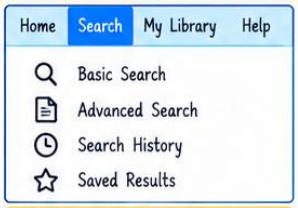
- The interface should be simple, clean and easy to understand.
- Use clear labels, intuitive icons and uncluttered layout.
- Avoid unnecessary complex options.
- Helps users, especially beginners, to search without confusion.



Simple and clear interface with essential options only

2. Consistency and Standards

- Use consistent layout, terminology, icons and colour scheme across all pages.
- Follow common standards and conventions (e.g., search icon 🔍, menu at top, similar button styles).
- Reduces learning time and helps users focus on their information need.



Consistent menu and icons across the interface

3. Flexibility and User Control

- Provide multiple ways to search (e.g., simple search, advanced search, filters, sorting).
- Allow users to modify their queries, refine results and control the search process.
- Support features like filter by date, author, document type, language etc.
- Helps both novice and expert users according to their needs.



Multiple options give control to the user

4. Feedback and Help

- Provide immediate and meaningful feedback for user actions.
- Show number of results, search time, suggestions for spelling correction, related terms etc.
- Offer help, tooltips and examples to guide users.
- Helps users understand the system and improve their search strategy.



Useful feedback, suggestions and help

5. Conclusion

- Human-Computer Interaction (HCI) focuses on designing systems that are easy, effective and user-friendly.
- Good interface design is essential in Information Retrieval systems to help users express their information needs and access relevant information.
- Design principles such as simplicity and clarity, consistency and standards, flexibility and user control, and feedback and help make search interfaces more usable and efficient.

Definition

Automatic Local Analysis is a technique used in an Information Retrieval (IR) system to automatically analyze the local context of words in a document or search query to identify important terms, phrases, and relationships.

It examines words that appear close to one another to understand their importance and improve searching and retrieval of relevant documents.

Example: In the sentence “Machine learning is used in medical image analysis,” the system can identify phrases such as machine learning and medical image analysis as important terms.

2. Working of Automatic Local Analysis

The process involves the following steps:

1. Input document: The system receives documents or search queries.
2. Local context analysis: It examines words within a small window around a particular word.
3. Feature extraction: It identifies frequently occurring terms, related words, and important phrases.
4. Indexing: The extracted terms and phrases are added to the document index.
5. Information retrieval: The system uses these features to improve search results.

3. Techniques Used in Automatic Local Analysis

1. Term Frequency Analysis

It identifies words that occur frequently within a local section of a document.

Example: In a paragraph about computers, the words “computer,” “software,” and “hardware” may occur frequently, indicating the main topic.

2. Co-occurrence Analysis

It identifies words that frequently appear near each other.

Example: In the phrase “Artificial Intelligence,” the words “Artificial” and “Intelligence” occur together and form a meaningful concept.

This helps the system recognize related terms and phrases.

3. N-gram Analysis

An N-gram is a sequence of \(N\) consecutive words or characters. It helps identify meaningful word combinations.

- Unigram: “data”
- Bigram: “data mining”
- Trigram: “data mining techniques”

Example: For the sentence “Data mining techniques are useful,” the system can identify “data mining” as a meaningful phrase.

4. Part-of-Speech (POS) Tagging

POS tagging identifies the grammatical role of each word, such as noun, verb, adjective, or adverb.

Example: In “Information retrieval system,” the system identifies “information,” “retrieval,” and “system” as nouns.

This helps identify important terms and noun phrases.

5. Keyphrase Extraction

It automatically identifies important phrases that represent the main topic of a document.

Example: From a research paper about detecting eye diseases using AI, the system may extract:

- Eye disease detection
- Artificial Intelligence
- Medical image analysis

These phrases can be used for indexing and searching.

4. Example of Automatic Local Analysis

Consider the following document:

“Machine learning is widely used in healthcare. Machine learning helps in disease prediction and medical image analysis.”

The system analyzes nearby words and identifies important terms and phrases.

Word or phrase	Analysis
Machine learning	Important technical phrase
Healthcare	Indicates the application area
Disease prediction	Meaningful two-word phrase
Medical image analysis	Important technical phrase

Result: The system can use these extracted terms to index the document. When a user searches for “disease prediction using machine learning,” the document may be retrieved because it contains related terms.

5. Applications in Information Retrieval

1. Query expansion: Suggests related terms to improve a user's search.
2. Document indexing: Identifies important words and phrases for faster retrieval.
3. Search result ranking: Helps identify terms that are relevant to a query.
4. Topic identification: Detects the main topics discussed in documents.
5. Document similarity: Helps identify documents containing similar terms and phrases.
6. Digital libraries: Helps users discover research papers related to their topics.

6. Advantages

- Automatically identifies important words and phrases.
- Improves the relevance of search results.
- Reduces the need for manual document analysis.
- Helps identify relationships between nearby words.
- Supports efficient indexing of large document collections.

7. Limitations

- May identify unimportant or irrelevant phrases.
- Results depend on the size of the local window and the analysis technique.
- May fail to understand relationships between words that are far apart.
- Requires additional processing time for large collections.

Q7. Discuss automatic local analysis, with suitable examples.

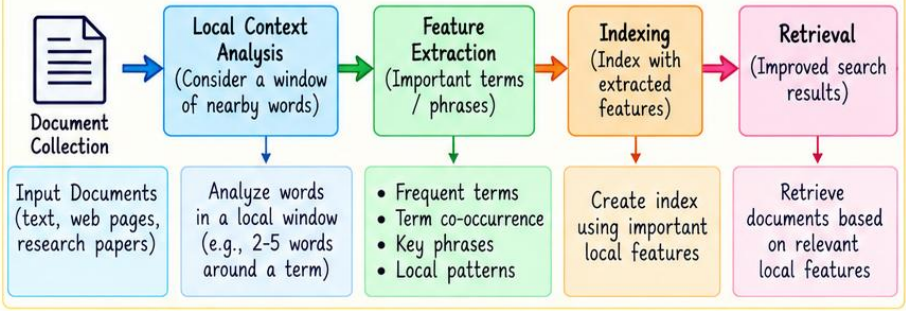
ANS

Automatic Local Analysis in Information Retrieval

1. Introduction

- Automatic local analysis is a technique used in Information Retrieval (IR) to automatically analyze the local context of documents and queries.
- It examines the words or terms that appear near each other (local windows) within a document or query to find important features, patterns or topics.
- The goal is to automatically identify significant terms or phrases that can improve indexing, search and retrieval of relevant documents.

2. Working of Automatic Local Analysis



3. Techniques Used in Automatic Local Analysis

1. Term Frequency Analysis – identifies frequently occurring terms in a local window.
2. Co-occurrence Analysis – finds terms that appear together in a local context.
3. N-gram Extraction – extracts meaningful phrases (e.g., bigrams, trigrams) from nearby words.
4. Part-of-Speech (POS) Tagging – identifies important nouns, noun phrases and key terms.
5. Named Entity Recognition (NER) – detects important entities like person names, organizations, locations.
6. Keyphrase Extraction – automatically selects relevant phrases from the local context.

4. Example 1: Term Co-occurrence and Keyphrase Extraction

Document: "Machine learning is widely used in healthcare. Machine learning helps in disease prediction and medical image analysis."

Local Window Analysis (window size = 2)

Term	Nearby Terms (Local Context)	Observation
machine	learning, is	Often occurs with "learning"
learning	machine, is, widely	Strong co-occurrence with "machine"
healthcare	in, widely	Related to "machine learning"
disease	in, prediction, and	Forms key phrase "disease prediction"
image	medical, analysis	Forms key phrase "medical image analysis"

Extracted Important Features

- machine learning
- healthcare
- disease prediction
- medical image analysis
- learning
- prediction
- image analysis

5. Example 2: N-gram Extraction

Text: "Data mining techniques are effective for big data analysis."

bigrams (2-grams)

- data mining
- mining techniques
- techniques are
- are effective
- effective for
- for big
- big data
- data analysis

trigrams (3-grams)

- data mining techniques
- mining techniques are
- techniques are effective
- are effective for
- effective for big
- for big data
- big data analysis

6. Applications in Information Retrieval

- Query expansion – suggests related terms based on local context.
- Improved indexing – index important phrases and entities.
- Document similarity – identifies similar documents using local patterns.
- Topic detection – discovers main topics in documents.
- Named entity search – helps in searching for persons, organizations or locations.
- Better ranking – improves relevance of retrieved documents.

7. Advantages

- Automatically finds important terms and phrases.
- Captures local context and meaning.
- Improves search accuracy and relevance.
- Helps in discovering hidden patterns.
- Useful for large document collections.

8. Limitations

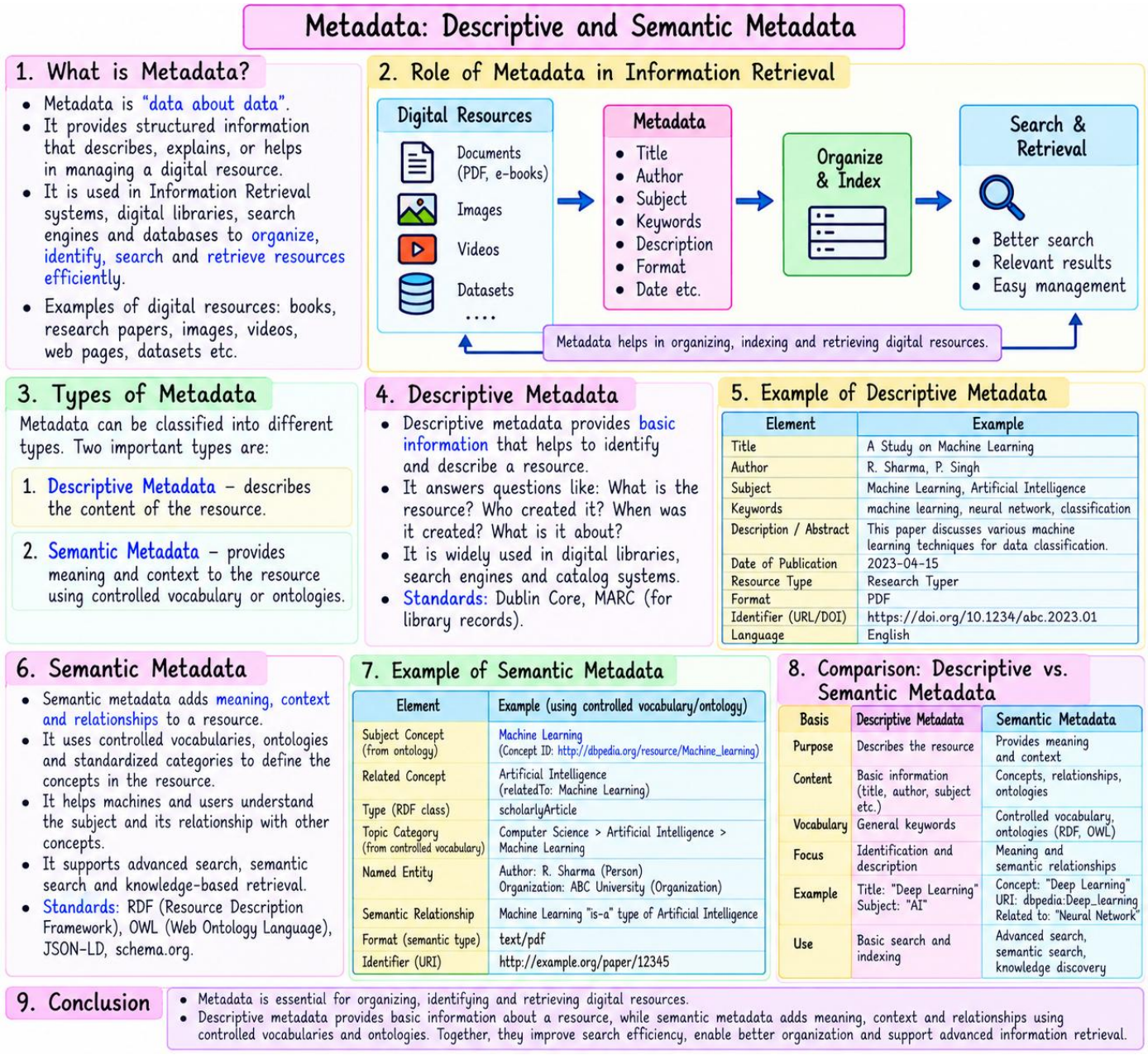
- May extract irrelevant or noisy terms.
- Depends on window size and parameters.
- Context may be lost for long-distance relationships.
- Requires additional processing time.

9. Conclusion

Automatic local analysis automatically examines the local context of words in documents and queries to identify important terms, phrases and patterns. It uses techniques such as term frequency, co-occurrence, n-grams, POS tagging and named entity recognition. By capturing local features, it improves indexing, search and retrieval of relevant documents in an Information Retrieval system.

Q8. What is Metadata? Explain Descriptive and Semantic Metadata in detail.

ANS



What is Metadata?

Metadata is defined as “data about data.” It provides information about a digital resource, such as its title, author, subject, date, format, and meaning.

Metadata helps users identify, organize, search, and retrieve information efficiently from digital libraries, databases, and search engines.

Example: For a research paper, its title, author's name, publication date, and keywords are metadata.

2. Types of Metadata

Two important types of metadata are:

1. Descriptive Metadata
2. Semantic Metadata

3. Descriptive Metadata

Definition: Descriptive metadata provides basic information that helps identify and describe a digital resource.

It helps users understand what a resource is, who created it, and what it contains.

Main elements of descriptive metadata

- Title: Name of the resource.
- Author: Name of the person who created it.
- Subject: Topic covered by the resource.
- Keywords: Important words related to the content.
- Description: Short summary of the resource.
- Date: Date of publication or creation.
- Format: File type, such as PDF, DOCX, or MP4.
- Identifier: A unique identifier, such as a DOI or document ID.

Example

Consider a research paper about Artificial Intelligence.

Metadata element	Example
Title	Introduction to Artificial Intelligence
Author	Rahul Sharma
Subject	Artificial Intelligence

Keywords	AI, Machine Learning, Robotics
Publication date	2025
Format	PDF

These details help users identify the required paper without reading the entire document.

Advantages of descriptive metadata

1. Makes documents easy to identify.
2. Helps users search for specific resources.
3. Supports document classification and indexing.
4. Helps manage digital library collections.
5. Saves time when searching for information.

Applications

Descriptive metadata is commonly used in digital libraries, online book catalogues, research paper repositories, and document management systems.

4. Semantic Metadata

Definition: Semantic metadata provides information about the meaning of a resource and its relationships with other concepts or resources.

It helps computers and users understand the context of information rather than only its basic details.

Semantic metadata may use controlled vocabularies, taxonomies, ontologies, and standardized relationships.

Main features of semantic metadata

- Meaning: Explains what a concept represents.
- Relationships: Connects related concepts.
- Controlled vocabulary: Uses standardized terms to avoid confusion.
- Context: Identifies the intended meaning of a word or topic.
- Ontologies: Define concepts and relationships in a particular domain.

Example

Consider a digital library containing a document about Machine Learning.

Its semantic metadata may include:

Element	Example
Main concept	Machine Learning
Broader concept	Artificial Intelligence
Related concept	Deep Learning
Relationship	Machine Learning is a branch of Artificial Intelligence
Application	Image Recognition

This information helps a search system connect related topics and retrieve relevant documents, even when users search using related concepts.

Advantages of semantic metadata

1. Improves the relevance of search results.
2. Helps identify relationships between concepts.
3. Supports searches using related terms and concepts.
4. Reduces ambiguity in the meaning of words.
5. Supports semantic search and knowledge-based systems.

Applications

Semantic metadata is used in knowledge graphs, semantic search engines, digital libraries, research databases, and intelligent information retrieval systems.

5. Difference Between Descriptive and Semantic Metadata

Basis	Descriptive Metadata	Semantic Metadata
Meaning	Describes a resource	Explains meaning and relationships
Purpose	Identifies the resource	Helps understand its context
Information	Title, author, date, keywords	Concepts, relationships, ontology terms
Focus	Identification and description	Meaning and connections

Example	Title: Machine Learning	Machine Learning is a branch of AI
Use	Basic search and indexing	Concept-based and advanced search

6. Role of Metadata in Information Retrieval

Metadata plays an important role in Information Retrieval Systems.

1. Indexing: Helps create searchable indexes for digital resources.
2. Searching: Allows users to search by title, author, subject, or keywords.
3. Filtering: Helps narrow results by date, author, format, or topic.
4. Ranking: Supports the selection and ranking of relevant documents.
5. Resource organization: Makes large digital collections easier to manage.
6. Improved discovery: Semantic relationships help users find related information.

UT-1

Q1. .What is the purpose of using keyword based query? Briefly explain any 3 types of keyword based queries.

ANS

Keyword-Based Queries in Information Retrieval System

1. Introduction

A **keyword-based query** is a type of query in which the user enters one or more **keywords or important terms** related to the required information. The Information Retrieval System (IRS) compares these keywords with the contents of documents in its database and retrieves the documents that are most relevant to the query.

Example:
If a user wants information about machine learning algorithms, the keyword-based query can be:

“Machine Learning Algorithms”

The IRS searches for documents containing these terms and ranks the results according to their relevance.

---

2. Purpose of Using Keyword-Based Queries

The main purposes are:

1. **Simple and easy searching:** Users can search information using a few meaningful words.
2. **Fast information retrieval:** The system can quickly match keywords with indexed documents.
3. **Improves relevance:** Important terms help the system identify documents related to the user's requirement.
4. **No complex query language required:** Users do not need to know database query syntax.
5. **Supports large collections:** Keyword searching is useful for searching large databases, websites, digital libraries, and search engines.
6. **Flexible searching:** Users can combine keywords to make the search more specific.

---

3. Types of Keyword-Based Queries

Any **three** of the following types can be explained in an examination.

A. Single Keyword Query

A **single keyword query** contains only one keyword or search term.

Example:

Blockchain

The IRS retrieves documents that contain or are associated with the word **Blockchain**.

Advantages:

- Very simple to enter.
- Useful for broad searches.
- Requires minimum effort from the user.

Limitation:
It may return a large number of irrelevant documents because the query is very broad.

B. Multiple Keyword Query

A **multiple keyword query** contains two or more keywords related to the information required.

Example:

Artificial Intelligence Machine Learning

The system searches for documents containing these keywords and retrieves documents that are related to the topic.

Advantages:

- More specific than a single keyword query.
- Reduces irrelevant results.
- Helps improve retrieval accuracy.

Example use:
A student searching for information about **IoT security** can use:

IoT Security Attacks

C. Boolean Keyword Query

A **Boolean keyword query** uses Boolean operators such as **AND, OR, and NOT** to combine keywords and control the search.

1. AND

Retrieves documents containing **both** keywords.

IoT AND Security

Result: Documents containing both **IoT** and **Security**.

2. OR

Retrieves documents containing **either or both** keywords.

AI OR Artificial Intelligence

Result: Documents containing either term.

3. NOT

Excludes documents containing a particular keyword.

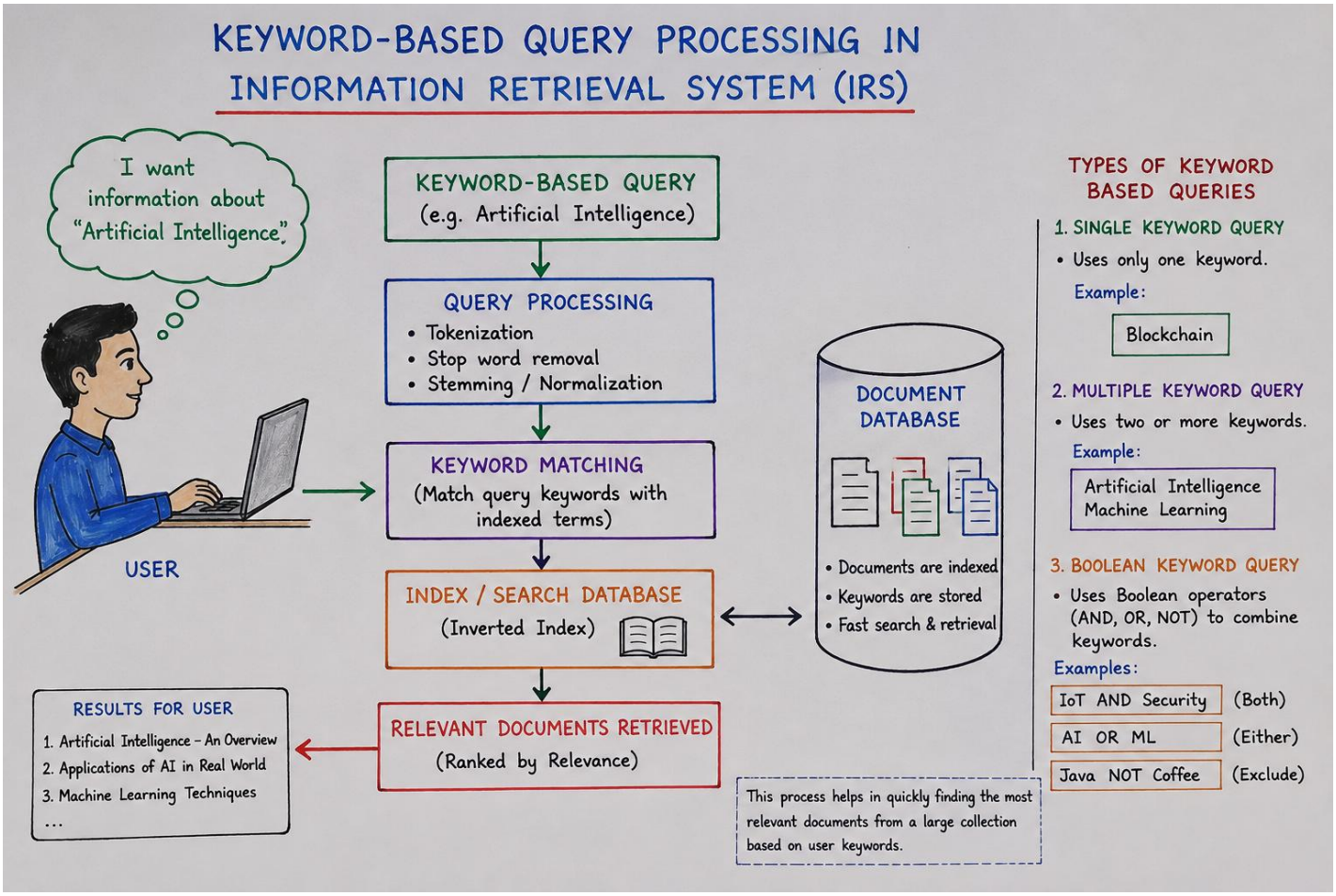
Java NOT Coffee

Result: Documents related to **Java** while excluding documents related to **Coffee**.

Advantage:
Boolean operators provide better control over the search and can improve the precision of retrieved results.

---

4. Diagram



5. Conclusion

Keyword-based queries provide a **simple, fast, and effective way to retrieve information** from an Information Retrieval System. They allow users to search using meaningful terms without requiring complex query languages. **Single keyword, multiple keyword, and Boolean keyword queries** are commonly used methods, with Boolean queries providing greater control over the search results.

Q2. Define Natural Language with features and applications

ANS

Definition of Natural Language

Natural Language refers to the language that humans naturally use to **communicate with each other**, both verbally and in written form. Examples include **English, Hindi, Marathi, French, etc.**

In an **Information Retrieval System (IRS)**, natural language allows users to express their information requirements in a human-understandable form. The system processes the user's natural-language query and retrieves relevant information from a collection of documents.

Example:
A user may enter:

"Find information about artificial intelligence applications in healthcare."

The IRS analyzes the words and meaning of the query and retrieves relevant documents.

---

2. Features of Natural Language

1. Easy and Human-Friendly

Natural language is easy for humans to understand and use. Users do not need to learn complex programming or query languages.

2. Ambiguity

A word or sentence can have **more than one meaning** depending on the context.

Example:
"Apple" can mean a **fruit** or the **technology company**.

3. Context Dependent

The meaning of a sentence often depends on the surrounding words and situation.

Example:
"Java is easy to learn" may refer to the **programming language**, while "Java is an island" refers to a geographical location.

4. Large Vocabulary

Natural languages contain a very large number of words, synonyms, abbreviations, and technical terms.

Example:
"Car", "automobile", and "vehicle" can have related meanings.

5. Flexibility

The same information can be expressed in different ways.

Example:

- "How can I learn Python?"
- "What is the best way to learn Python?"

Both can represent a similar information requirement.

6. Synonymy

Different words may have the same or similar meanings.

Example:
"Doctor" and "Physician".

An IRS should consider such relationships while retrieving documents.

7. Polysemy

A single word can have multiple related meanings.

Example:
The word **"bank"** may refer to a financial institution or the side of a river.

8. Grammatical Structure

Natural language follows grammatical rules involving nouns, verbs, adjectives, sentences, etc. Understanding this structure helps systems analyze queries correctly.

---

3. Applications of Natural Language

1. Information Retrieval

Natural language is widely used in search engines and IRS to allow users to search for information using ordinary sentences or keywords.

Example:

"Find books about machine learning."

---

2. Search Engines

Google and other search engines process natural-language queries and return relevant web pages.

Example:

“What is artificial intelligence?”

---

3. Question Answering Systems

Natural language is used by systems that understand questions and provide appropriate answers.

Example:

“Who invented the telephone?”

---

4. Chatbots and Virtual Assistants

Chatbots and assistants such as customer-support bots use natural language to communicate with users.

Example:

“What is my order status?”

---

5. Machine Translation

Natural language processing is used to translate text from one language to another.

Example:

English → Hindi
“Good morning” → “सुप्रभात”

---

6. Speech Recognition

Natural language processing helps convert human speech into text and understand spoken commands.

Example:

“Set an alarm for 7 AM.”

---

7. Text Summarization

Natural language techniques can identify important information from large documents and generate a shorter summary.

Example:
Summarizing a long research paper into its main points.

---

8. Sentiment Analysis

Natural language is used to determine whether a text expresses a **positive, negative, or neutral** opinion.

Example:

“The product is excellent.” → Positive sentiment.

---

4. Natural Language Processing in IRS

NATURAL LANGUAGE PROCESSING IN INFORMATION RETRIEVAL SYSTEM (IRS)



USER

Find information about artificial intelligence applications in healthcare.

NATURAL LANGUAGE QUERY

User enters query in natural language.

QUERY PROCESSING

- Tokenization
- Stop-word Removal
- Stemming
- NLP Analysis (Parsing, POS Tagging, etc.)

QUERY–DOCUMENT MATCHING

Processed query is matched with indexed documents in the system.

DOCUMENT COLLECTION (INDEX / DATABASE)

- Documents are stored and indexed
- Keywords / terms are extracted and stored
- Enables fast search and retrieval

RELEVANT DOCUMENTS RETRIEVED

System retrieves and ranks documents based on relevance.

RESULTS

Relevant documents are displayed to the user.

5. Conclusion

Natural Language is the **human-friendly form of communication** used to express information and queries. In an Information Retrieval System, it helps users search for information without requiring knowledge of complex query languages. Its features such as **ambiguity, context dependency, large vocabulary, flexibility, synonymy, and grammatical structure** make natural-language processing important for modern IRS applications such as **search engines, chatbots, question-answering systems, machine translation, speech recognition, and text summarization**.

Q3. Define and Discuss Objective of IRS

ANS

Definition of Information Retrieval System

An **Information Retrieval System (IRS)** is a computer-based system used to **store, organize, search, and retrieve relevant information** from a large collection of documents according to the user's information requirement.

The information may be in the form of **text documents, web pages, research papers, books, images, emails, or other digital content**.

Example:
When a student searches for **"Applications of Artificial Intelligence in Healthcare"**, an IRS searches its document collection and returns the most relevant documents.

Simple Definition:

An Information Retrieval System is a system that retrieves relevant information from a large collection of information resources based on a user's query.

---

2. Objectives of Information Retrieval System

The major objectives of an IRS are as follows:

1. Retrieve Relevant Information

The primary objective is to retrieve documents that are **relevant to the user's query**.

For example, for the query:

Machine Learning Algorithms

the system should return documents related to machine learning algorithms rather than unrelated documents.

---

2. Provide Fast Retrieval

An IRS should provide search results **quickly**, even when the collection contains millions of documents.

Search indexes such as an **inverted index** are used to make retrieval faster.

---

3. Improve Precision

Precision refers to the proportion of retrieved documents that are actually relevant to the user's query.

The system aims to **minimize irrelevant results** and provide highly useful documents.

Example:
If 10 documents are retrieved and 8 are relevant, precision is:

Precision = 8/10 = 0.8 or 80%

---

4. Improve Recall

Recall refers to the ability of the system to retrieve **as many relevant documents as possible** from the entire collection.

A good IRS attempts to retrieve important relevant documents without missing them.

Example:
If there are 20 relevant documents in the database and the system retrieves 15 of them:

Recall = 15/20 = 75%

---

5. Handle Large Volumes of Information

Modern information collections contain huge amounts of data. An IRS should efficiently manage and search **large document collections**.

Examples include:

- Digital libraries
- Search engines
- Research databases
- Online repositories

---

6. Understand User Queries

The system should process and understand different types of user queries, including **keywords, phrases, Boolean queries, and natural-language queries**.

For example:

AI AND Healthcare

The system should correctly interpret the query and retrieve suitable documents.

---

7. Rank Retrieved Documents

An IRS should **rank documents according to their relevance** so that the most useful results appear at the top.

For example:

Query: "Artificial Intelligence"

↓

1. AI Applications in Healthcare ← Highly Relevant
2. Introduction to Artificial AI ← Relevant
3. History of Computer Science ← Less Relevant

---

8. Provide Easy Access to Information

The system should provide a **simple and user-friendly interface** through which users can search and access information easily.

Users should not need advanced technical knowledge to use the system.

---

9. Reduce Information Overload

The amount of information available on the Internet and in databases is extremely large. IRS helps users **filter unnecessary information** and find useful information efficiently.

---

10. Support Different Types of Information

An IRS should be capable of retrieving information from different sources and formats such as:

- Text
- Web pages
- Books
- Research papers
- Images
- Emails
- Multimedia

---

3. Diagram of Objectives of IRS

OBJECTIVES OF INFORMATION RETRIEVAL SYSTEM (IRS)

INFORMATION RETRIEVAL SYSTEM

USER QUERY
(Information need expressed as words/keywords)



OBJECTIVES
1. Retrieve Relevant Information
2. Provide Fast Retrieval
3. Improve Precision
4. Improve Recall
5. Handle Large Volumes of Information
6. Understand User Queries
7. Rank Retrieved Documents
8. Provide Easy Access
9. Reduce Information Overload
10. Support Different Types of Information

QUERY PROCESSING
- Analysis of query
- Stop words removal
- Stemming / Normalization

SEARCH / INDEXING
- Search the indexed database
- Match query with documents
- Find relevant documents

DOCUMENT COLLECTION / DATABASE
(Indexed Documents)

FAST RETRIEVAL
Results are provided in minimum time.

HIGH PRECISION
Retrieve more relevant documents and minimize irrelevant results.

HIGH RECALL
Retrieve as many relevant documents as possible.

UNDERSTAND USER QUERIES
Process different types of queries (keywords, Boolean, Natural language etc.)

RELEVANCE RANKING
Rank documents according to their relevance.

RELEVANT INFORMATION TO USER
Most useful and relevant documents are displayed to the user.

USER
(Satisfied with relevant information)

4. Importance of IRS

An Information Retrieval System is important because it:

1. Saves **time and effort**.
2. Provides **quick access** to required information.
3. Helps users search **large databases**.
4. Reduces irrelevant information.
5. Improves the **accuracy and relevance** of search results.
6. Supports decision-making and research.
7. Makes information easily accessible to users.

Q4. Differentiate befween Data Retrieval and Information Retrieval.

ANS

Difference Between Data Retrieval and Information Retrieval

Basis	Data Retrieval	Information Retrieval
1. Meaning	Retrieves exact data from a database.	Retrieves relevant information from a collection of documents.
2. Data Type	Mainly works with **structured data**.	Mainly works with **unstructured or semi-structured data**.
3. Query	Uses precise and well-defined queries.	Can use keywords, phrases, Boolean queries, or natural language.
4. Query Language	Commonly uses **SQL** and database query languages.	Uses search queries and IR techniques.
5. Matching	Usually requires an **exact match** based on conditions.	Uses **relevance-based matching**.
6. Result	Produces exact records or values.	Produces a ranked list of relevant documents/information.
7. Ranking	Usually does not rank results based on relevance.	Results are commonly ranked according to relevance.
8. Precision & Recall	Precision and recall are generally not the main evaluation measures.	**Precision and recall** are important evaluation measures.
9. Uncertainty	Results are generally deterministic.	Results may contain uncertainty because relevance can vary.
10. Example	Finding students with marks > 80.	Finding research papers about AI in healthcare.

Q5. State the different types of queries. Explain the pattern matching query concept with an example.

ANS

1. Introduction

A **query** is a request submitted by a user to an Information Retrieval System to find required information from a collection of documents.

An IRS supports different types of queries depending on how the user wants to search for information.

---

2. Different Types of Queries

The important types of queries in an Information Retrieval System are:

1. Keyword-Based Query

A keyword-based query consists of one or more important words related to the required information.

Example:

Artificial Intelligence Healthcare

The system searches for documents containing these keywords.

---

2. Boolean Query

A Boolean query uses logical operators such as **AND, OR, and NOT**.

Example:

AI AND Healthcare

It retrieves documents containing both **AI** and **Healthcare**.

Other operators:

- **AND** → Both terms must be present.
- **OR** → Either term or both may be present.
- **NOT** → Excludes a particular term.

---

3. Phrase Query

A phrase query searches for an **exact sequence of words**.

Example:

"Information Retrieval System"

The system searches for documents containing the complete phrase in the same order.

---

4. Pattern Matching Query

A pattern matching query searches for words or terms that **match a specified pattern**. It is useful when the user knows only part of a word or wants to search for words having a common pattern.

Example:

comput*

This may retrieve:

- computer
- computing
- computation
- computational

Pattern matching commonly uses **wildcards** such as * and ?.

---

5. Natural Language Query

A natural language query is expressed in the same way that people normally communicate.

Example:

“What are the applications of AI in healthcare?”

The IRS processes the query and retrieves relevant documents.

---

6. Proximity Query

A proximity query searches for terms that occur **near each other** within a document.

Example:

machine NEAR learning

It retrieves documents where the words **machine** and **learning** occur close to each other.

---

3. Pattern Matching Query

Definition

A **Pattern Matching Query** is a query in which the user specifies a **pattern of characters or words**, and the Information Retrieval System retrieves terms or documents that match that pattern.

It is particularly useful when:

- The exact spelling of a word is unknown.
- The user wants to search for different forms of a word.
- A word has different prefixes or suffixes.
- The user wants to retrieve multiple related terms at once.

Pattern matching is generally implemented using **wildcard characters**.

---

4. Wildcard Characters

A. Asterisk (*)

The **asterisk (*)** represents **zero or more characters**.

Example:

comput*

It can match:

- computer
- computing
- computation
- computational

Another example:

bio*

It may match:

- biology
- biological
- biotechnology
- bioinformatics

B. Question Mark (?)

The **question mark (?)** generally represents **a single character**.

Example:

wom?n

It can match:

- woman
- women

The exact wildcard syntax may vary between different search systems.

---

5. Example of Pattern Matching

Suppose an IRS contains the following words:

computer

computing

computation

computational

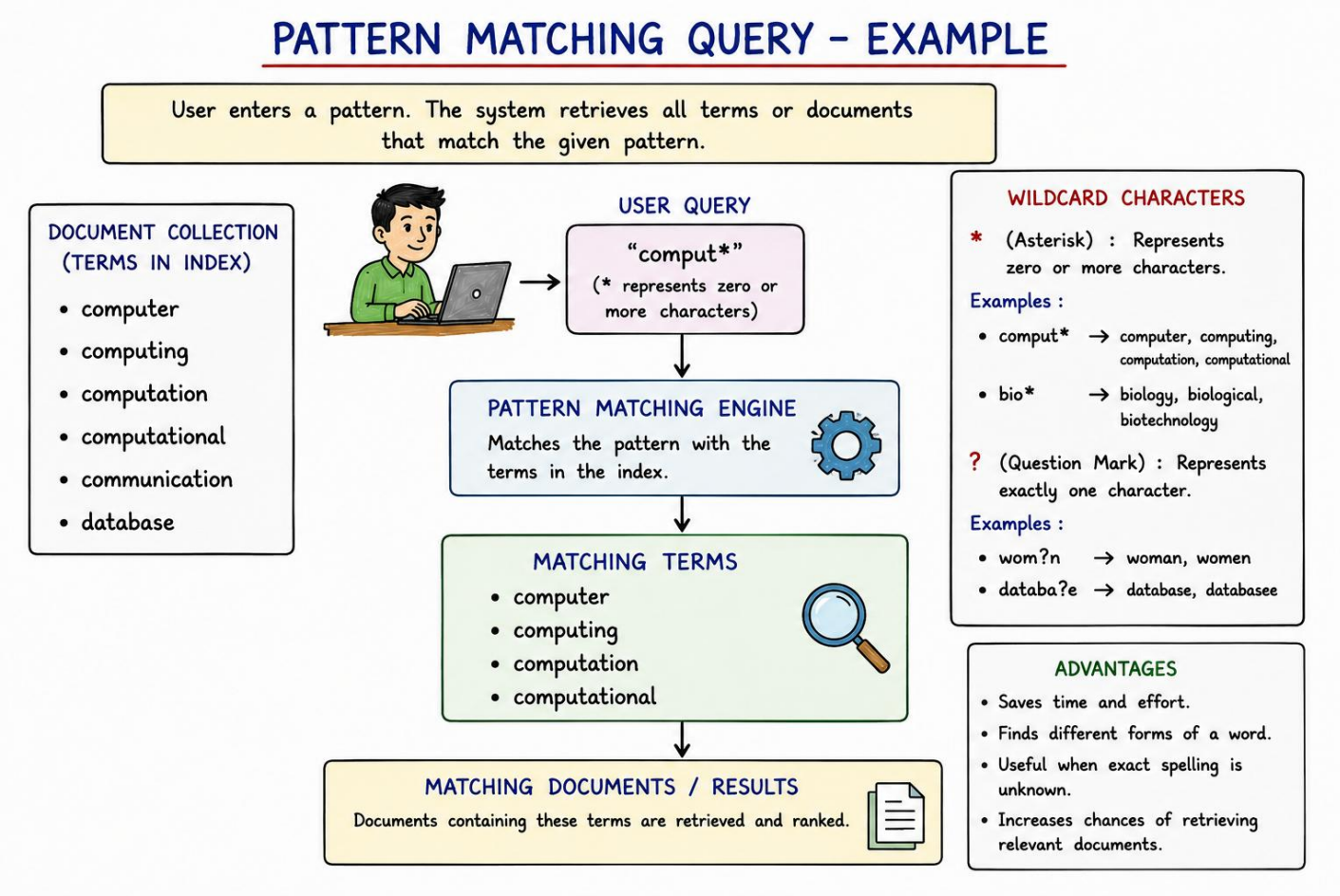
communication

database

User enters:

comput*

The system identifies all terms beginning with **comput**.



6. Advantages of Pattern Matching Queries

1. **Saves time** because multiple terms can be searched using one query.
2. Helps find **different forms of a word**.
3. Useful when the user is **not sure about the complete spelling**.
4. Increases the possibility of finding relevant documents.
5. Useful for searching **large document collections**.
6. Makes searching more **flexible**.
7. Reduces the need to enter multiple similar keywords.

7. Limitations

1. A broad pattern may retrieve **too many documents**.
2. It can increase the number of **irrelevant results**.
3. Different search systems may support different wildcard rules.
4. Complex patterns may increase the **search processing time**.

Q6. Explain taxonomy of information retrieval model with classification Diagram.

ANS

TAXONOMY OF INFORMATION RETRIEVAL MODEL

1. INTRODUCTION

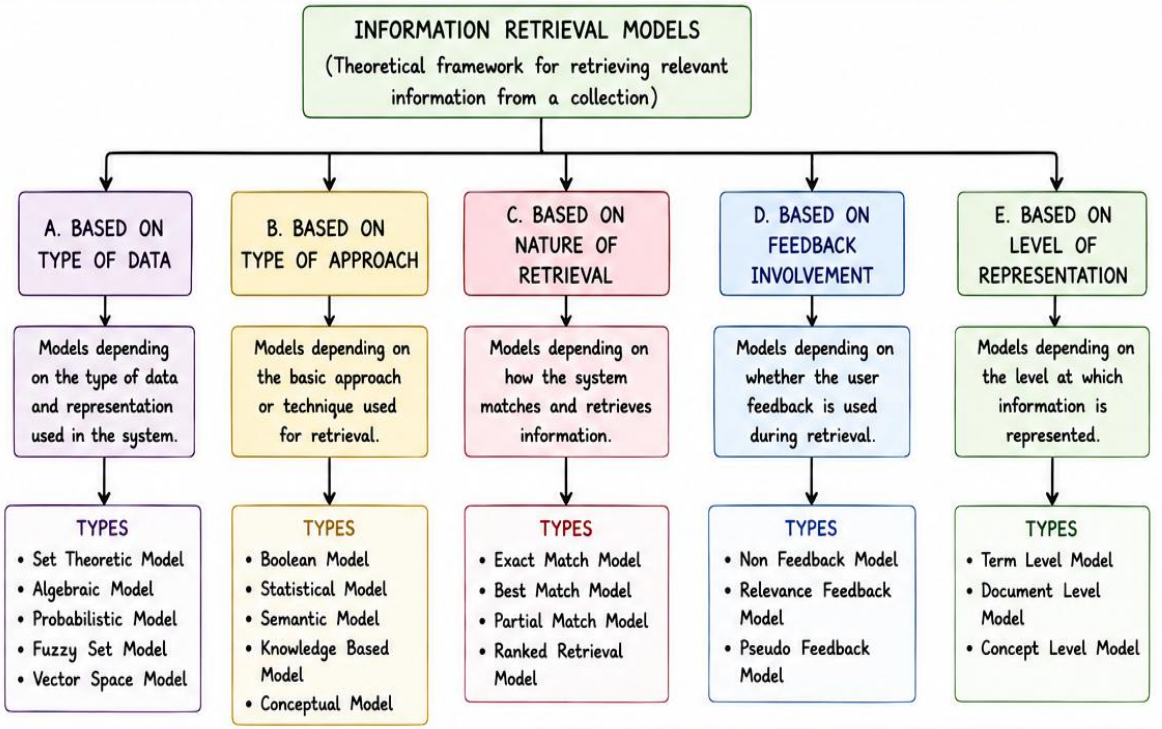
An Information Retrieval (IR) Model is a theoretical framework used to represent, organize and retrieve information from a collection of documents in response to a user query.

2. TAXONOMY (CLASSIFICATION)

IR models can be classified on the basis of various parameters such as the type of data used, the technique used and the nature of retrieval.

BASIS OF CLASSIFICATION

- Type of data used
- Type of approach
- Nature of retrieval
- Feedback involvement
- Level of representation



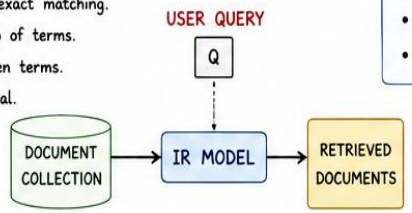
3. EXPLANATION OF MAJOR MODELS

1. Set Theoretic Model : Represents documents and queries as sets of terms.
2. Vector Space Model : Represents documents and queries as vectors in a multi-dimensional space.
3. Probabilistic Model : Retrieves documents based on probability of relevance.
4. Boolean Model : Uses Boolean logic (AND, OR, NOT) for exact matching.
5. Fuzzy Set Model : Handles vagueness and partial membership of terms.
6. Semantic Model : Considers meaning and relationship between terms.
7. Knowledge Based Model : Uses domain knowledge to improve retrieval.

EXAMPLE OF CLASSIFICATION

Query : "Data Mining in Healthcare"

- In Boolean Model (Data AND Mining AND Healthcare)
- In Vector Space Model → Vector representation and cosine similarity
- In Probabilistic Model → Documents with highest probability are retrieved
- In Relevance Feedback → Initial results → User feedback → Improved results



General working of an IR Model

Note:
In practice, many modern IR systems use a combination of models to achieve better retrieval performance.

4. CONCLUSION

Different IR models provide different ways to represent documents and queries and to compute relevance. The taxonomy helps us choose the most suitable model based on the requirements of the system.

Introduction

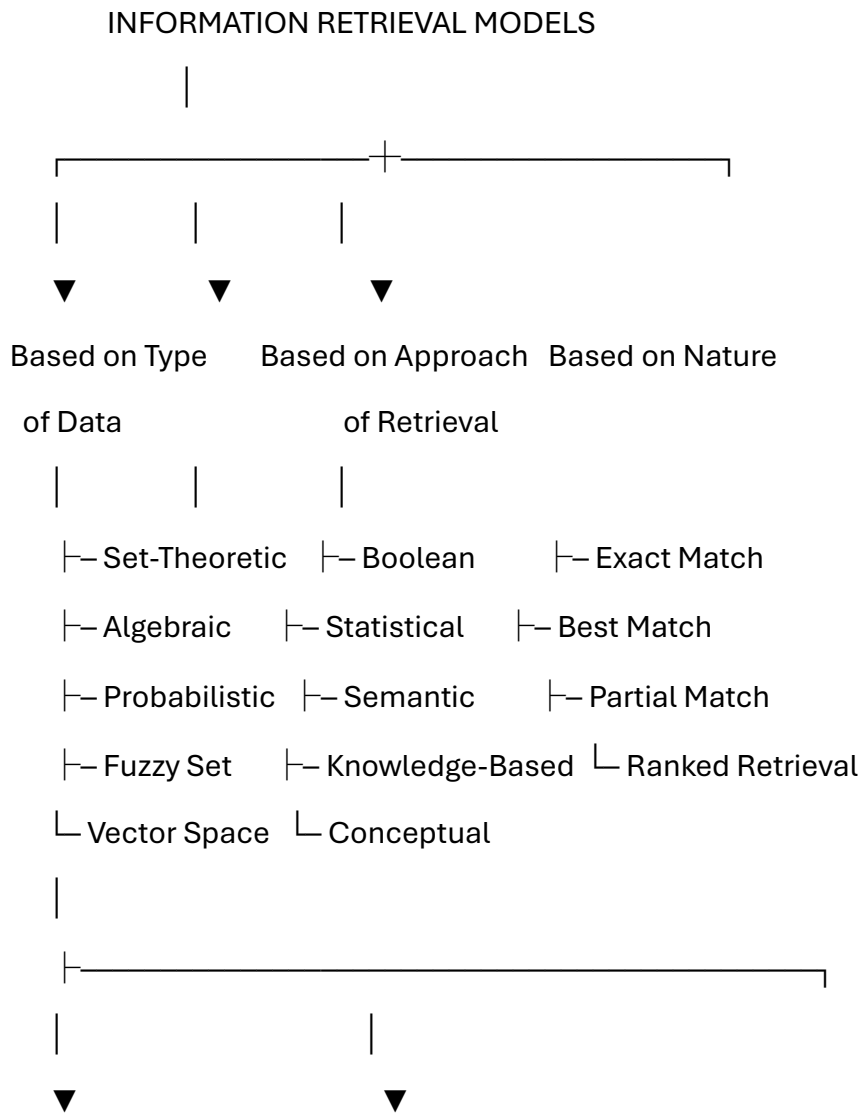
An **Information Retrieval (IR) Model** is a theoretical framework used to **represent documents and user queries, compare them, and retrieve relevant information** from a collection of documents.

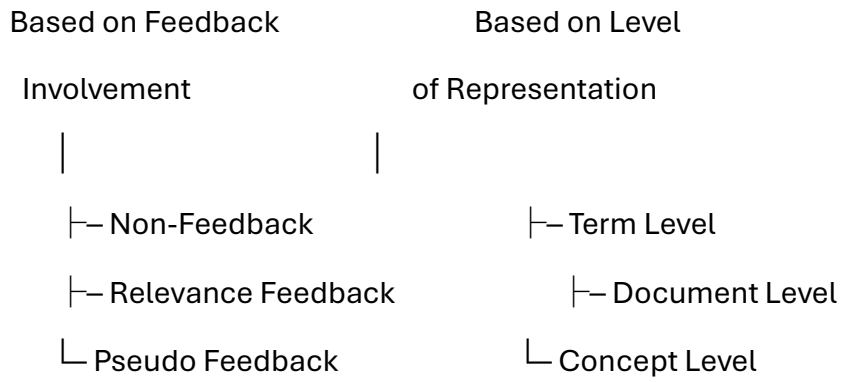
The taxonomy of IR models classifies different models according to the **type of data representation, retrieval approach, matching method, feedback mechanism, and level of representation**.

---

2. Classification / Taxonomy of IR Models

The major classification can be represented as:





---

3. Major Types of IR Models

A. Set-Theoretic Models

These models represent documents and queries as **sets of terms**.

1. Boolean Model

It uses Boolean operators such as:

- **AND**
- **OR**
- **NOT**

Example:

AI AND Healthcare

Documents containing both terms are retrieved.

2. Fuzzy Set Model

It extends the Boolean model by allowing **partial membership** instead of only true or false matching.

B. Algebraic Models

These models represent documents and queries using mathematical structures.

Vector Space Model

Documents and queries are represented as **vectors in a multidimensional space**.

Similarity between a query and document can be calculated using **cosine similarity**.

Example:

Query: Machine Learning

Documents containing more relevant terms receive higher similarity scores.

C. Probabilistic Models

The **Probabilistic Model** estimates the probability that a document is relevant to a particular query.

Documents are ranked according to their **probability of relevance**.

Example:

If Document A has a higher probability of being relevant than Document B, Document A is displayed first.

D. Semantic Models

Semantic models consider the **meaning and relationships between words**, rather than only matching exact terms.

Example:

A query containing:

automobile

may also retrieve documents containing:

car

because they have related meanings.

E. Knowledge-Based Models

These models use **domain knowledge, concepts, relationships, and rules** to improve retrieval.

They are useful in specialized domains such as:

- Medical information retrieval
- Legal information systems
- Scientific databases

---

4. Classification Based on Nature of Retrieval

1. Exact Match Model

Retrieves documents that satisfy the query exactly.

2. Best Match Model

Retrieves documents that are **most similar or relevant** to the query.

3. Partial Match Model

Retrieves documents that partially satisfy the user's query.

4. Ranked Retrieval Model

Documents are assigned relevance scores and displayed from **most relevant to least relevant**.

---

5. Classification Based on Feedback

1. Non-Feedback Model

The system retrieves documents without receiving additional information from the user.

2. Relevance Feedback Model

The user identifies which retrieved documents are relevant, and the system uses this feedback to improve subsequent results.

3. Pseudo-Relevance Feedback

The system automatically assumes that the top retrieved documents are relevant and uses them to improve the query.

---

6. Classification Based on Level of Representation

1. Term-Level Model

Represents information using individual terms or keywords.

2. Document-Level Model

Considers the document as a complete information unit.

3. Concept-Level Model

Represents information using concepts and their relationships.

---

7. Important IR Models for Examination

Model	Main Idea
Boolean Model	Uses AND, OR, NOT operators
Vector Space Model	Represents documents as vectors and calculates similarity
Probabilistic Model	Estimates probability of document relevance
Fuzzy Set Model	Allows partial membership
Semantic Model	Considers meaning of terms
Knowledge-Based Model	Uses domain knowledge and relationships

8. Example

Consider the query:

“Data Mining in Healthcare”

Different models process it differently:

- **Boolean Model:** Data AND Mining AND Healthcare
- **Vector Space Model:** Represents query and documents as vectors and calculates similarity.
- **Probabilistic Model:** Ranks documents according to their probability of relevance.
- **Semantic Model:** May consider related concepts such as *medical analytics* and *healthcare data analysis*.
- **Relevance Feedback:** Uses the user's feedback to improve future results.**of an IR Model**

10. Conclusion

The **taxonomy of Information Retrieval models** provides a systematic way to classify IR models according to how they **represent documents and queries, perform matching, calculate relevance, use feedback, and represent information**.

Q7. Explain Probabilistic Rerieval Model. Compare and Contrast Belief Network Model and Interference Network Model.

ANS

1. Probabilistic Retrieval Model

Definition

The **Probabilistic Retrieval Model** is an Information Retrieval model based on the idea that a document is retrieved according to the **probability that it is relevant to a given user query**.

The model attempts to estimate:

P(Relevant | Document, Query)

and ranks documents in **decreasing order of their probability of relevance**.

The basic assumption is that for every query, there is a set of documents that is relevant and another set that is not relevant.

---

2. Working of Probabilistic Retrieval Model

The working can be explained in the following steps:

Step 1: User submits a query

The user enters a query containing keywords.

Example:

Machine Learning in Healthcare

Step 2: Initial retrieval

The system searches the document collection and identifies potentially relevant documents.

Step 3: Probability estimation

The system estimates the probability that each document is relevant to the query.

Step 4: Document ranking

Documents are ranked according to their estimated probability of relevance.

Step 5: Results are displayed

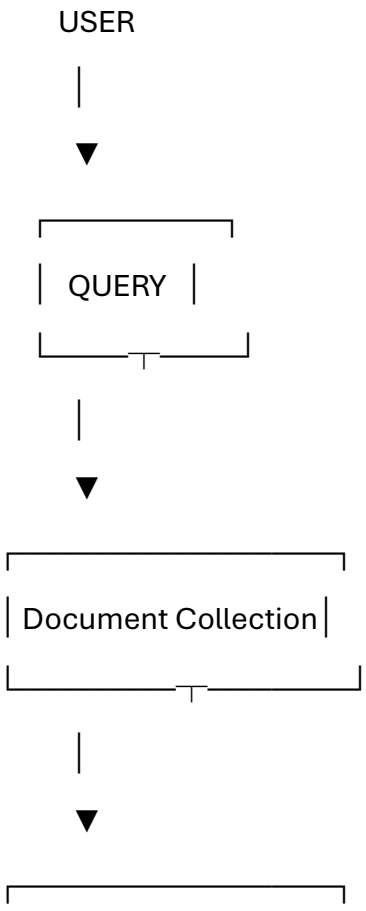
The documents with the highest probability of relevance are displayed first.

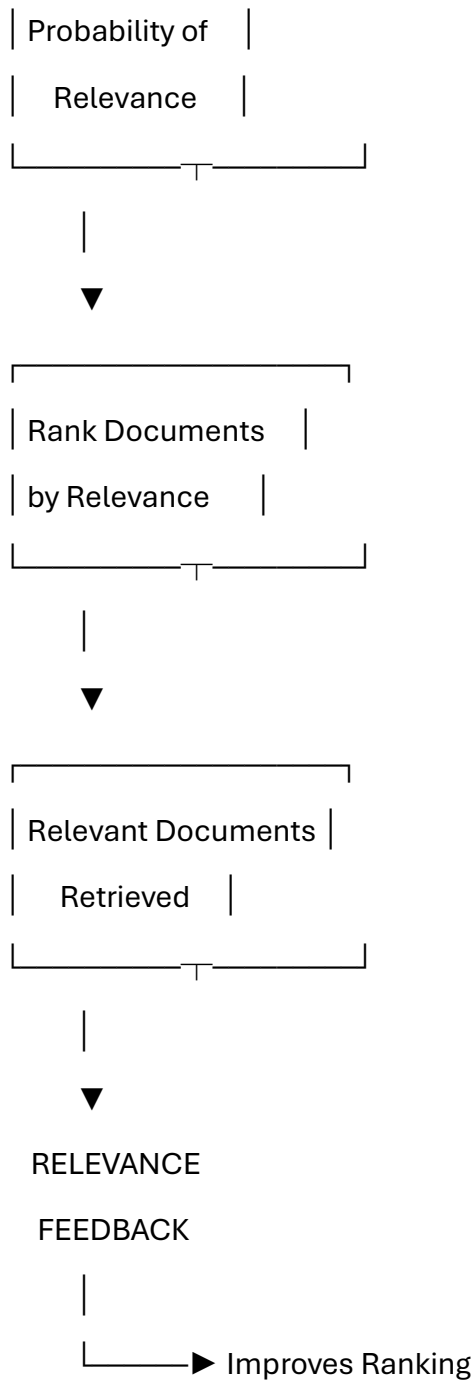
Step 6: Relevance feedback

The user may identify relevant documents. This feedback can be used to improve the probability estimates and retrieve better results.

---

3. Diagram of Probabilistic Retrieval Model





---

4. Advantages of Probabilistic Retrieval Model

1. **Ranks documents according to relevance probability.**
2. Provides a natural framework for **relevance-based retrieval**.
3. Supports **relevance feedback**.

4. Gives better ranking than simple exact matching.
5. Can handle uncertainty in document relevance.
6. Useful for large document collections and search systems.

---

5. Limitations

1. Initial probability estimates may not always be accurate.
2. It may require relevance information or feedback.
3. Estimating probabilities can be computationally complex.
4. The model may assume independence between terms in some implementations.
5. Performance depends on the quality of the statistical estimates.

---

6. Belief Network Model

A **Belief Network Model** uses a **Bayesian/probabilistic network** to represent relationships and dependencies between different concepts, terms, documents, and queries.

The network consists of **nodes and directed links**.

- **Nodes** represent concepts, terms, documents, or queries.
- **Links** represent relationships or dependencies.
- Probabilities associated with the network help determine the likelihood of relevance.

It is useful when relationships between terms are important.

---

7. Inference Network Model

The **Inference Network Model** represents information retrieval as a process of **inference**.

It consists of a network through which evidence from documents is propagated toward a query node.

A typical inference network contains:

- **Document representation**
- **Term/concept nodes**
- **Query representation**
- **Inference relationships**
- **Query evaluation**

The system estimates how strongly the available document evidence supports a user's query and ranks documents accordingly.

---

8. Belief Network vs Inference Network

Basis	Belief Network Model	Inference Network Model
1. Basic idea	Represents probabilistic relationships among concepts/terms.	Represents retrieval as an inference process between documents and queries.
2. Main focus	Modeling **dependencies and beliefs**.	Modeling **evidence and inference** for retrieval.
3. Structure	Uses a Bayesian-style directed network.	Uses a network connecting document representations, concepts, and query representations.
4. Representation	Represents probabilistic relationships among variables.	Represents how evidence from documents supports a query.
5. Probability	Probability is used to represent beliefs and dependencies.	Probability is used to estimate the strength of inference/relevance.
6. Query processing	Query concepts are evaluated using the probability relationships in the network.	Query nodes receive evidence propagated from document nodes.
7. Main objective	Determine the likelihood/belief associated with concepts or documents.	Determine the probability or degree to which a document satisfies the query.

Basis	Belief Network Model	Inference Network Model
8. Dependencies	Explicitly models dependencies between variables.	Focuses more on evidence propagation and inference.
9. Retrieval	Retrieves documents based on probabilistic belief.	Ranks documents based on the strength of inference supporting the query.
10. Use	Useful when relationships among concepts are important.	Useful for flexible query evaluation and probabilistic information retrieval.

10. Key Difference to Remember

Belief Network

Focuses on probabilistic relationships and dependencies among concepts or variables.

Inference Network

Focuses on propagating evidence through a network to determine how strongly a document supports a query.

1. PROBABILISTIC RETRIEVAL MODEL

Definition :

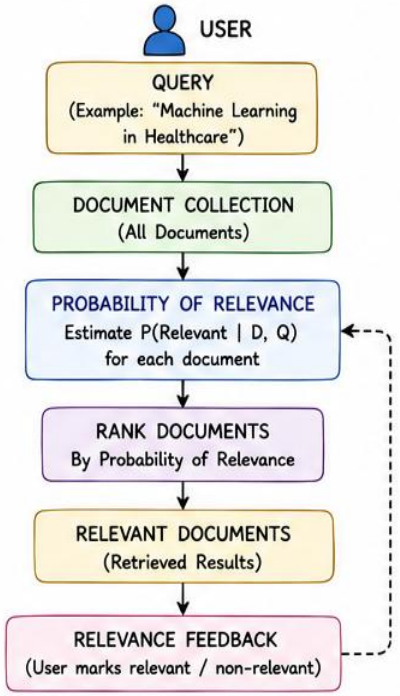
The Probabilistic Retrieval Model is based on the idea that a document is retrieved according to the probability that it is relevant to a given user query.

Main Idea :

Documents are ranked in decreasing order of P(Relevant | Document, Query)

Working Steps :

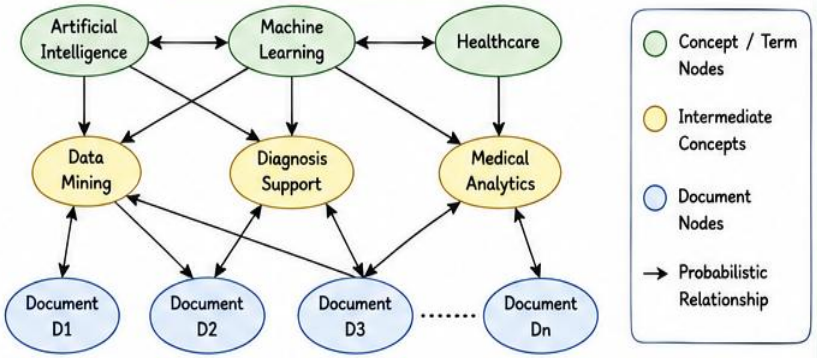
1. User submits a query.
2. Initial retrieval from document collection.
3. Estimate the probability of relevance for each document.
4. Rank documents by probability of relevance.
5. Display top ranked documents.
6. Relevance feedback (optional) is used to improve further results.



2. COMPARISON OF BELIEF NETWORK AND INFERENCE NETWORK

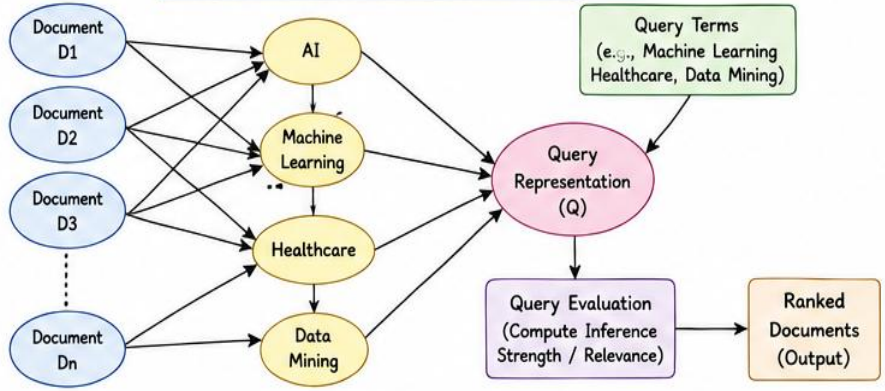
Sr. No.	Basis	Belief Network Model	Inference Network Model
1.	Basic Idea	Represents probabilistic relationships among concepts / terms.	Represents retrieval as an inference process between documents and queries.
2.	Main Focus	Modeling dependencies and beliefs.	Modeling evidence and inference for retrieval.
3.	Structure	Uses a Bayesian-style directed network.	Uses a network connecting documents, concepts and query representations.
4.	Representation	Represents probabilistic relationships among variables.	Represents how evidence from documents supports a query.
5.	Probability	Probability represents beliefs and dependencies.	Probability represents strength of inference / relevance.
6.	Query Processing	Query concepts are evaluated using the probability relationships in network.	Query nodes receive evidence propagated from document nodes.
7.	Main Objective	Determine the likelihood / belief associated with concepts or documents.	Determine the probability or degree to which a document satisfies the query.
8.	Dependencies	Explicitly models dependencies between variables.	Focuses more on evidence propagation and inference.
9.	Retrieval	Retrieves documents based on probabilistic belief.	Ranks documents based on the strength of inference supporting the query.
10.	Use	Useful when relationships among concepts are important.	Useful for flexible query evaluation and probabilistic information retrieval.

3. BELIEF NETWORK MODEL



Belief Network represents probabilistic relationships among concepts.
Probabilities on links indicate strength of belief / dependency.

4. INFERENCE NETWORK MODEL



Evidence from documents is propagated through the network to the query node.
Strength of inference determines the relevance of each document.

Q8. Explain Classic Information Retrieval models.

ANS

CLASSIC INFORMATION RETRIEVAL MODELS

1. BOOLEAN MODEL

Documents and queries are represented as sets of terms using Boolean logic.

Boolean Operators
AND OR NOT

Example Query
AI AND Healthcare

Matching
Exact / Binary Matching (0 or 1)

Result
Documents either satisfy the query or do not satisfy the query.

2. VECTOR SPACE MODEL

Documents and queries are represented as vectors in a multi-dimensional space.

Term Weighting
TF-IDF (Term Frequency – Inverse Document Frequency)

Similarity Measure
Cosine Similarity

$$\text{Cosine}(D, Q) = \frac{D \cdot Q}{|D||Q|}$$

Result
Documents are ranked based on their similarity score with the query.

3. PROBABILISTIC MODEL

Documents are ranked based on the probability of being relevant to the query.

Probability Estimation
Estimate P(Relevant | Document, Query)

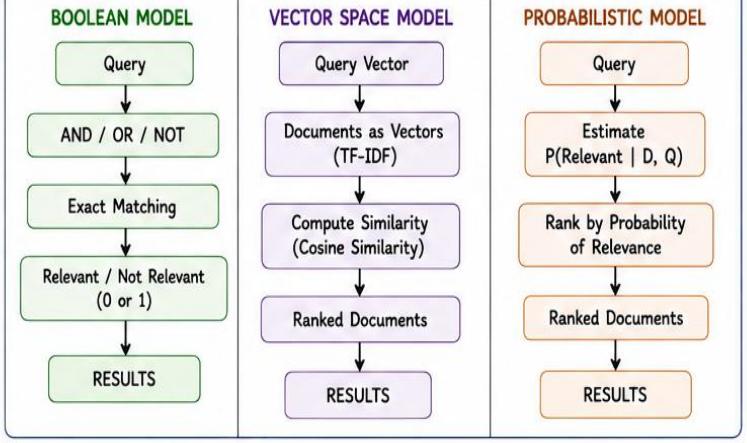
Ranking
Documents are ranked in decreasing order of probability of relevance.

Result
Documents with higher probability are shown first.

COMPARISON OF CLASSIC IR MODELS

Feature	Boolean Model	Vector Space Model	Probabilistic Model
Representation	Set of terms	Vector of terms	Probability of relevance
Matching	Exact / Boolean matching	Similarity matching	Probability-based matching
Main Technique	AND, OR, NOT	TF-IDF, Cosine Similarity	Probability estimation
Ranking	No (Binary result)	Yes (Similarity score)	Yes (Probability score)
Partial Matching	Limited	Yes	Yes
Relevance	Binary (0 or 1)	Similarity value	Probability value
Feedback	Limited	Can be supported	Strongly supported
Main Advantage	Simple, precise control	Effective ranked retrieval	Relevance-based ranking
Main Limitation	No ranking, vague results	No semantic meaning, large vector	Probability estimation is complex

DIAGRAM OF CLASSIC IR MODELS



KEY POINTS: • Boolean Model – Uses logical operators; exact matching. • Vector Space Model – Uses term weights and similarity; provides ranking. • Probabilistic Model – Uses probability of relevance; best for ranked retrieval.

1. Introduction

An **Information Retrieval (IR) Model** is a mathematical or theoretical framework used to represent **documents and user queries** and determine the relevance of documents to a given query.

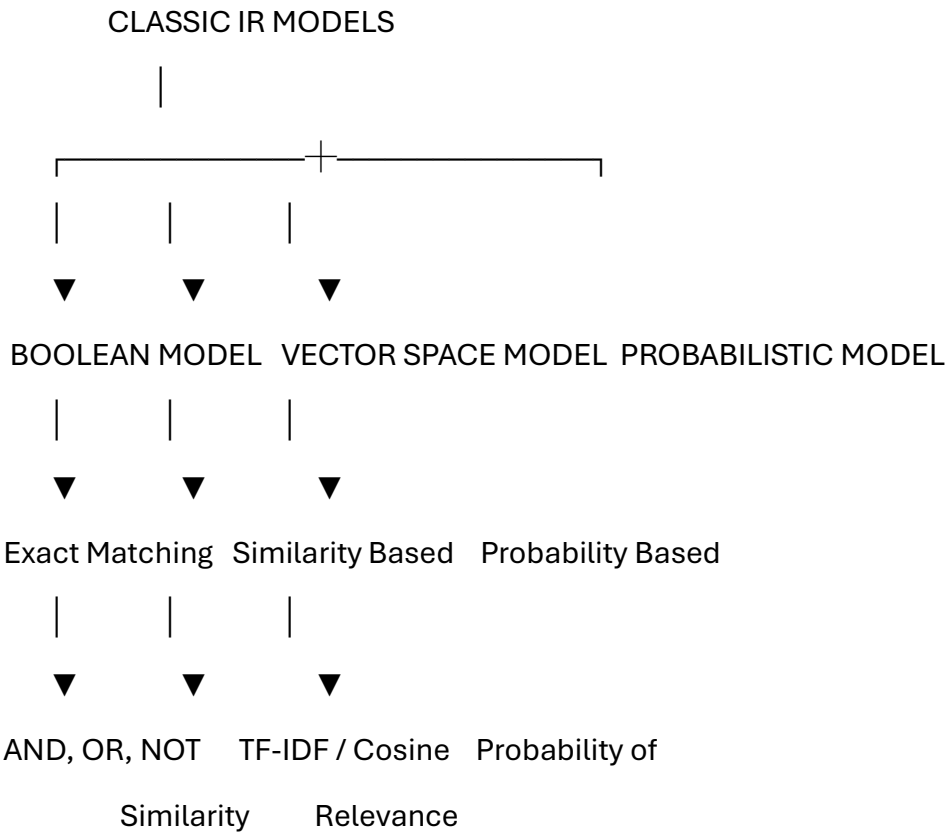
The **Classic Information Retrieval Models** are the fundamental models on which many modern retrieval techniques are based.

The three major classic IR models are:

1. **Boolean Model**
2. **Vector Space Model**
3. **Probabilistic Model**

---

2. Classification of Classic IR Models



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3. Boolean Model

Definition

The **Boolean Model** is a classical information retrieval model in which documents and queries are represented using **sets of terms**. It uses Boolean operators such as:

- **AND**
- **OR**
- **NOT**

A document either **satisfies the query or does not satisfy it**.

Example

Query:

AI AND Healthcare

The system retrieves documents containing **both AI and Healthcare**.

Another example:

AI OR Machine Learning

The system retrieves documents containing either or both terms.

Advantages

- Simple and easy to understand.
- Supports precise query formulation.
- Useful for database and library searching.
- Allows users to control the search using Boolean operators.

Limitations

- Does not provide a natural ranking of documents.
- Results are generally either relevant or non-relevant.
- Users must formulate queries carefully.
- It may return either too many or too few documents.

---

4. Vector Space Model

Definition

The **Vector Space Model (VSM)** represents documents and queries as **vectors in a multidimensional space**.

Each dimension represents a term, and the weight of a term indicates its importance in a document.

The model commonly uses **TF-IDF (Term Frequency–Inverse Document Frequency)** to calculate term weights.

Working

1. Documents are converted into vectors.
2. The user's query is also converted into a vector.
3. The similarity between the query and each document is calculated.
4. Documents are ranked according to their similarity.

Cosine Similarity

A common similarity measure is:

A higher cosine similarity indicates that the document is more similar to the query.

Example

Query:

Machine Learning

Suppose three documents have similarity scores:

Document	Similarity
D1	0.92
D2	0.75
D3	0.31

The retrieval order will be:

D1 → D2 → D3

Advantages

- Provides **ranked retrieval**.
- Supports partial matching.
- Easy to calculate similarity between documents and queries.
- More flexible than the Boolean model.

Limitations

- Does not directly capture the semantic meaning of words.
- Synonyms may be treated as different terms.
- Term independence is generally assumed.
- Vector representation can become very large for large collections.

---

5. Probabilistic Model

Definition

The **Probabilistic Model** ranks documents according to the **probability that they are relevant to a user's query**.

The basic idea is to estimate:

Documents having a higher probability of relevance are ranked higher.

Working

1. User submits a query.
2. The system identifies candidate documents.
3. It estimates the probability of relevance for each document.
4. Documents are ranked according to probability.
5. Relevant documents are presented to the user.
6. **Relevance feedback** can be used to improve subsequent retrieval.

Example

For the query:

Artificial Intelligence Healthcare

Suppose:

Document	Probability of Relevance
D1	0.90
D2	0.78
D3	0.42

The system displays:

D1 → D2 → D3

Advantages

- Provides ranked retrieval.
- Naturally handles uncertainty.
- Supports relevance feedback.
- Focuses directly on document relevance.

Limitations

- Initial probability estimates may not be accurate.
- Probability calculation can be complex.
- Performance depends on the quality of available information and estimates.

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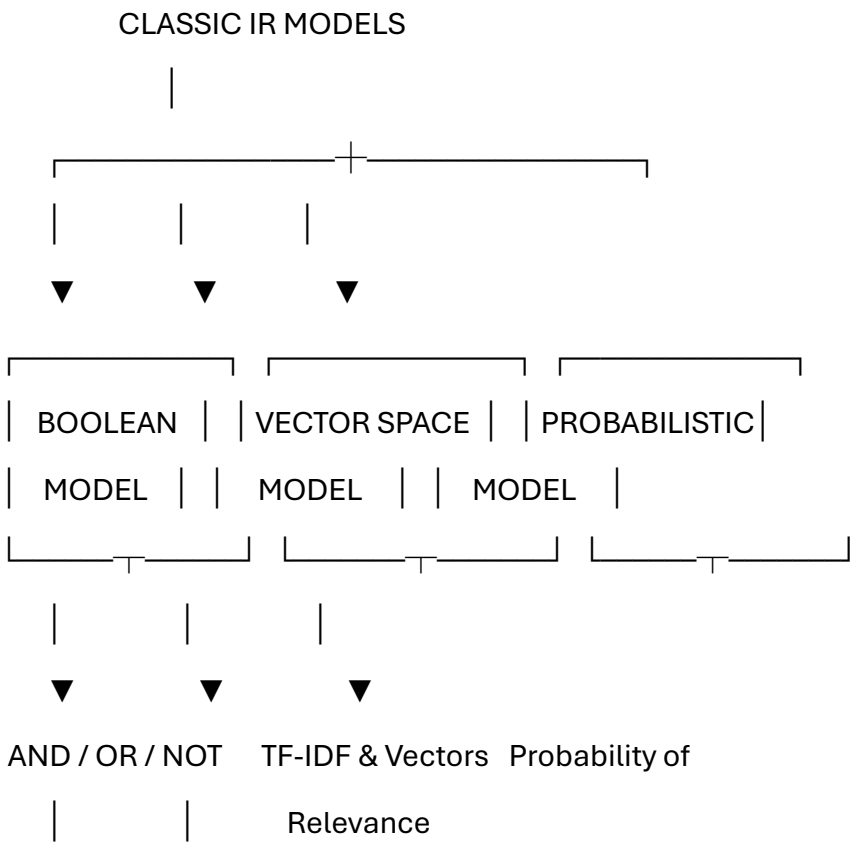
6. Comparison of Classic IR Models

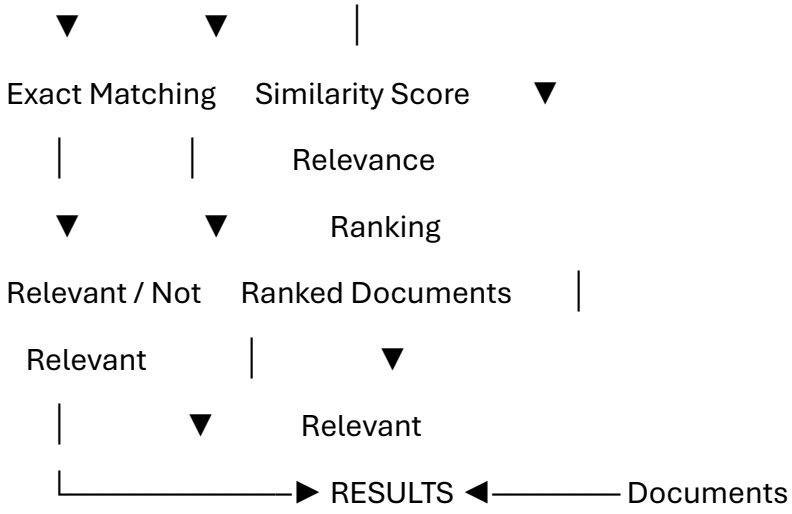
Feature	Boolean Model	Vector Space Model	Probabilistic Model
Representation	Sets of terms	Vectors	Probability of relevance
Matching	Exact/Boolean matching	Similarity matching	Probability-based matching
Main operators/technique	AND, OR, NOT	TF-IDF, Cosine Similarity	Probability estimation

Feature	Boolean Model	Vector Space Model	Probabilistic Model
Ranking	Generally No	Yes	Yes
Partial matching	Limited	Yes	Yes
Relevance	Binary	Similarity score	Probability
Query complexity	Can be complex	Relatively simple	Relatively simple
Feedback	Limited	Can be supported	Strongly supported
Main advantage	Simple and precise control	Effective ranked retrieval	Relevance-based ranking
Main limitation	No natural ranking	Limited semantic understanding	Probability estimation can be difficult

---

7. Diagram of Classic IR Models





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8. Key Points to Remember for Exam

Boolean Model

Set-based model → AND, OR, NOT → exact/binary matching

Vector Space Model

Vector-based model → TF-IDF → cosine similarity → ranked results

Probabilistic Model

Probability-based model → probability of relevance → ranked results

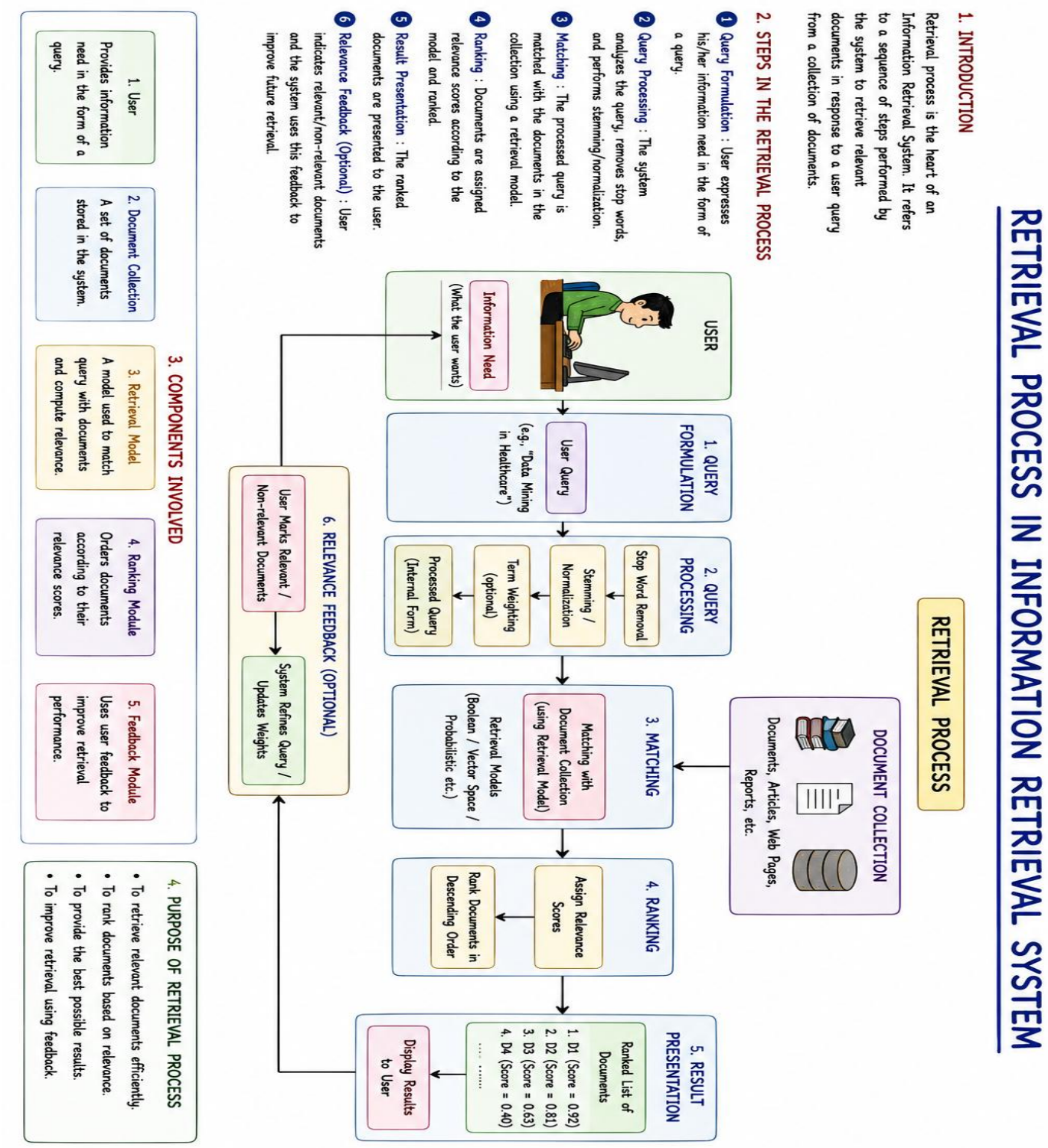
---

9. Conclusion

The **Boolean, Vector Space, and Probabilistic models** are the three important **Classic Information Retrieval Models**. The Boolean model uses exact logical matching, the Vector Space Model uses **term weights and similarity measures** to rank documents, and the Probabilistic Model ranks documents according to their **probability of relevance**. These models form the foundation of many advanced Information Retrieval techniques used in **search engines, digital libraries, research databases, and document retrieval systems**.

Q9. Explain the Retrieval Process with neat labelled diagram.

ANS



1. Definition

The **retrieval process** is the sequence of steps performed by an Information Retrieval System to **find, select, rank, and present relevant documents** from a large collection according to the user's information need or query.

The main goal is:

To retrieve the most relevant information for the user in the shortest possible time.

---

2. Steps in the Retrieval Process

1. Information Need

The process starts when a user has a requirement for some information.

Example:
A student wants information about:

"Applications of Data Mining in Healthcare."

The user's information need is converted into a searchable query.

---

2. Query Formulation

The user expresses the information requirement as a **query** using keywords, phrases, Boolean operators, or natural language.

Example:

Data Mining AND Healthcare

---

3. Query Processing

The system processes the user's query before searching the document collection.

Common operations include:

- **Tokenization** – Breaking the query into individual terms.
- **Stop-word removal** – Removing common words such as *the, is, a, an*.
- **Stemming** – Reducing words to their root form.

- **Normalization** – Converting terms into a standard form.
- **Term weighting** – Assigning importance to terms when required.

The result is a **processed query**.

---

4. Document Collection and Indexing

The IRS contains a collection of documents such as:

- Books
- Research papers
- Web pages
- Articles
- Reports
- Emails

These documents are **indexed** so that the system can search them efficiently.

An **inverted index** is commonly used for fast searching.

---

5. Matching / Retrieval

The processed query is compared with the indexed documents using an **IR model**.

Common models include:

- Boolean Model
- Vector Space Model
- Probabilistic Model

The system identifies documents that are potentially relevant to the query.

---

6. Ranking

The retrieved documents are assigned **relevance scores**.

Documents are then arranged from **most relevant to least relevant**.

For example:

Document	Relevance Score
D1	0.92
D2	0.81
D3	0.63
D4	0.40

Therefore:

D1 → D2 → D3 → D4

---

7. Result Presentation

The ranked results are presented to the user through the search interface.

The system may display:

- Document title
- Short description/snippet
- URL or document location
- Relevance/ranking information

---

8. Relevance Feedback

Relevance feedback is an optional step in which the user indicates which retrieved documents are relevant or non-relevant.

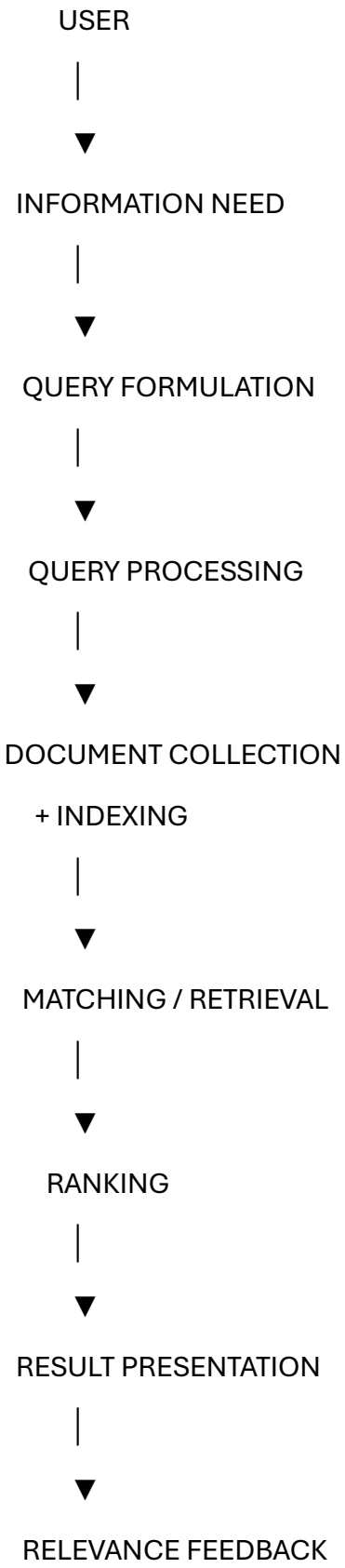
The system uses this information to **refine the query and improve future retrieval results**.

---

3. Retrieval Process Diagram

The **neat labelled diagram is provided above in image format** for direct use in your exam preparation.

Simplified flow:


USER
INFORMATION NEED
QUERY FORMULATION
QUERY PROCESSING
DOCUMENT COLLECTION
+ INDEXING
MATCHING / RETRIEVAL
RANKING
RESULT PRESENTATION
RELEVANCE FEEDBACK



---

4. Components Involved in Retrieval

1. User

Provides the information requirement or query.

2. Document Collection

Contains documents from which information is retrieved.

3. Query Processor

Analyzes and transforms the user's query.

4. Retrieval Model

Matches the query with documents and determines their relevance.

5. Ranking Module

Ranks retrieved documents according to relevance scores.

6. Feedback Module

Uses user feedback to improve retrieval performance.

---

5. Purpose of the Retrieval Process

The retrieval process aims to:

1. Retrieve **relevant documents** efficiently.
2. Reduce the number of **irrelevant results**.
3. Provide **fast information access**.
4. Rank documents according to their **relevance**.
5. Improve retrieval using **user feedback**.
6. Provide the **best possible results** to the user.

---

6. Example

Suppose the user searches:

“Artificial Intelligence in Healthcare”

The retrieval process works as:

User Query → Query Processing → Index Search → Document Matching → Relevance Scoring → Ranking → Results

The system may return:

1. *Applications of AI in Healthcare* — 0.95
2. *AI-Based Medical Diagnosis* — 0.87
3. *Machine Learning in Healthcare* — 0.78

The most relevant document is displayed first.

---

7. Conclusion

The **retrieval process is the core of an Information Retrieval System**. It starts with the user's information need and proceeds through **query formulation, query processing, indexing, matching, ranking, and result presentation**. Relevance feedback can further improve the quality of retrieval. The overall objective is to provide the **most relevant information quickly and efficiently**.

Q10. Compare and contrast Proximity Queries and Wildcard Queries

ANS

1. Introduction

Proximity Queries and **Wildcard Queries** are two important query techniques used in an Information Retrieval System. Both help users retrieve relevant information, but they solve different search problems.

- **Proximity Query:** Searches for terms that occur **near each other** in a document.
- **Wildcard Query:** Searches for terms that **match a specified character pattern**.

---

2. Proximity Query

Definition

A **Proximity Query** retrieves documents in which two or more specified terms occur **within a certain distance or position of each other**.

It is useful when the relationship between the words is important.

Example:

machine NEAR learning

This retrieves documents where **machine** and **learning** occur close to each other.

Another example:

"information" NEAR/3 "retrieval"

This can retrieve documents where the words **information** and **retrieval** occur within approximately three words of each other, depending on the search system's syntax.

Applications

- Searching phrases or related concepts.
- Legal document retrieval.
- Medical information retrieval.
- Searching research papers.
- Finding terms occurring close together.

---

3. Wildcard Query

Definition

A **Wildcard Query** uses special characters called **wildcards** to represent unknown or variable characters in a search term.

Common wildcard symbols include:

- * → represents **zero or more characters**
- ? → generally represents **one character**

Example 1:

comput*

May retrieve:

- computer
- computing
- computation
- computational

Example 2:

wom?n

May retrieve:

- woman
- women

Applications

- Searching different forms of a word.
- Searching when the exact spelling is unknown.
- Finding words with common prefixes or suffixes.
- Searching large document collections.

---

4. Difference Between Proximity and Wildcard Queries

Basis	Proximity Query	Wildcard Query
1. Definition	Searches for terms occurring close to each other.	Searches for terms matching a character pattern.
2. Main Purpose	Finds words based on their **position/distance**.	Finds words based on their **spelling/pattern**.
3. Focus	Relationship between search terms.	Character variation within a term.

Basis	Proximity Query	Wildcard Query
4. Query Example	machine NEAR learning	comput*
5. Main Operators	NEAR, WITHIN, proximity distance, etc.	*, ?
6. Matching	Matches terms according to their location in a document.	Matches terms according to their character pattern.
7. Example Result	machine learning, machine-based learning	computer, computing, computation
8. Order/Distance	Distance between terms is important.	Position of the word in the document is not the main concern.
9. Usefulness	Useful for phrases and related terms.	Useful for word variations and incomplete terms.
10. Main Limitation	Requires appropriate distance/operator settings.	Broad patterns may produce many irrelevant results.

---

5. Examples

A. Proximity Query

Suppose a document contains:

“Machine learning is an important branch of artificial intelligence.”

Query:

machine NEAR learning

Result: Document is retrieved because the terms **machine** and **learning** occur close to each other.

B. Wildcard Query

Suppose the index contains:

computer, computing, computation, computational, database

Query:

comput*

Result:

computer
computing
computation
computational

The * represents zero or more characters.

---

6. Diagram

7. Advantages

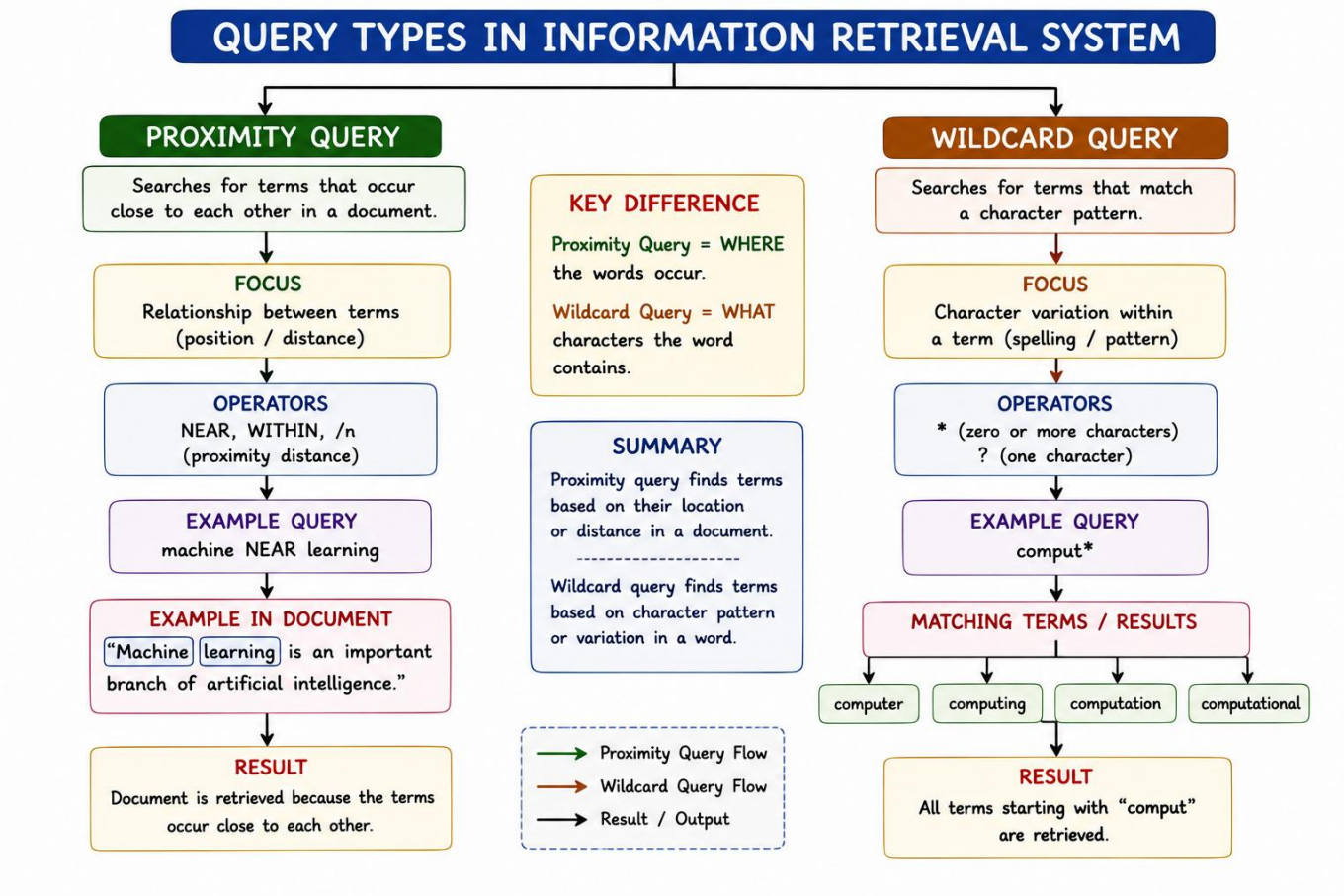
Proximity Query

1. Improves search precision.
2. Finds terms that are contextually related.
3. Useful for technical phrases.
4. Reduces irrelevant results.
5. Useful in legal, medical, and academic searching.

Wildcard Query

1. Finds multiple forms of a word.
2. Useful when spelling is incomplete or uncertain.
3. Saves time by avoiding multiple searches.
4. Provides flexible searching.
5. Useful for large document collections.

DIAGRAM



The easiest way to remember the difference is:

Proximity Query = WHERE the words occur.

Wildcard Query = WHAT characters the word contains.

Example:

Proximity:
data NEAR mining → searches for **data** and **mining** occurring close together.

Wildcard:
data* → searches for words beginning with **data**, such as *database* and *data mining* depending on the indexing/search system.

---

9. Conclusion

Proximity and Wildcard Queries improve the flexibility and effectiveness of Information Retrieval Systems. **Proximity Queries** focus on the **location and distance between terms**, while **Wildcard Queries** focus on **variations in the characters of a search term**. Proximity queries are useful for finding related terms or phrases, whereas wildcard queries are useful for finding different word forms and incomplete terms. Both techniques help users obtain more useful search results from large document collections.